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A Traceable Software Workflow for Supporting Scenario to Simulator Configuration

Design and Evaluation with CHESSCON

Pathipati Thanu Reddy

Faculty of Computing, Blekinge Institute of Technology, 371 79 Karlskrona, Sweden

This thesis is submitted to the Department of Software Engineering, Faculty of Computing, at Blekinge Institute of Technology in partial fulfilment of the requirements for the degree of Master of Science in Software Engineering. The thesis is equivalent to 20 weeks of full-time study.

The author declares sole authorship of this thesis and confirms that the sources used are identified in the text and bibliography. The author further declares that this thesis has not been submitted at another institution to obtain a degree.

This thesis was conducted with support from akquinet and the CHESSCON team, who provided the simulation context, practical guidance and access to the software used in the study. Permission was provided to reproduce the CHESSCON screenshots. Access to and redistribution of the underlying licensed projects remain subject to the permissions of the relevant owners.

Contact Information:
Author: Pathipati Thanu Reddy
E-mail: thpa24@student.bth.se

Academic supervisor:
Dr. Lawrence Henesey
Department of Computer Science, Blekinge Institute of Technology
E-mail: LHE@BTH.SE

Industry support:
Norbert Klettner
akquinet port consulting GmbH
E-mail: Norbert.Klettner@akquinet.de

Abstract

Background. Simulation studies normally begin with a written scenario describing what should be represented. However, simulation software requires this information to be entered through product-specific objects, settings and result files. A simulation may run successfully even when the sources of its settings, the decisions used during configuration and the effects of later changes are difficult to identify.
Objective. This thesis designs and evaluates a traceable, human-operated software workflow for organising information used in scenario-to-simulator configuration. The workflow supports configuration review, verification, evidence handling and change analysis around an existing CHESSCON model. Recent research is also used to define a baseline workflow for comparison. The study concerns the organisation of simulation work rather than terminal performance.
Method. The research combined a focused literature review, Design Science Research and a controlled comparison. Design Science Research develops and evaluates a practical solution to an identified problem. The review informed six baseline stages and cross-cutting information responsibilities. The proposed workflow was implemented through a workflow-support package containing controlled records, mappings, trace links, predefined checks, configuration guidance, change-impact records and an evidence catalogue. Both trials began from separate setups derived from an already configured reference model for the Electric Forklift Movement Scenario. The comparison did not evaluate construction from an empty project. One accepted execution per workflow was assessed using the same eight acceptance checks; full directory identity at accepted launch was not claimed.
Results. Both workflows produced accepted executions that met the same selected workload, equipment and charging conditions. The proposed workflow made the relationships between controlled information, simulator settings and checks explicit, and recorded the dependencies affected by later changes. Every selected information record had an internal trace, although coverage of all original source information was not established. The retained audit documented both satisfied checks and known interpretation issues. Workload and fleet changes were supported by executed results; the distance change was examined through impact analysis. Preparation was shorter in the recorded proposed-workflow observation, but incomplete timing and possible learning effects prevent a conclusion that it reduced overall modelling effort.
Conclusion. The proposed workflow supplied explicit information structures that were absent from the baseline materials while both accepted executions satisfied the same selected acceptance conditions. The study demonstrates recorded support for tracing and reviewing information, not complete scenario conversion, comparative usability, automation, productivity improvement or better simulator performance.

Keywords: software workflow; scenario-to-simulator configuration; requirements traceability; configuration management; simulation verification

Data Availability Statement

This thesis includes the derived tables, charts, screenshots and evidence summaries needed to understand the evaluation. The CHESSCON simulation results and supporting evidence were presented to the university supervisor during supervision.

The underlying CHESSCON projects and raw project files are licensed or controlled by the participating organisation and software provider. Therefore, these original project files are not distributed with the thesis. The results reported here are based on the preserved evaluation evidence and the documented analysis presented in the following chapters and appendix.

The thesis publicly presents the package structure, selected record and mapping examples, and derived evaluation results. The complete controlled records, mappings, trace table, check register and evidence catalogue remain in the author's private working files and are not supplied as a public supplementary archive. An examiner would need separate authorisation for any inspection of these retained files; this thesis does not establish that such access has been granted. The submitted PDF does not enable independent inspection of every underlying trace. Supervisor review does not establish public redistribution rights.

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A master's thesis may look like an individual achievement, but behind these pages is a rather wonderful team of people who helped make it possible.

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This thesis carries my name, but reaching this point would not have been the same without all of you.

And finally, to this thesis: you have taught me a great deal, but I think we are both ready for me to graduate.

Declaration of Generative AI Usage

With reference to the GenAI Policy for course PA2534 at Blekinge Institute of Technology, the generative AI assistance used during the preparation of this thesis is disclosed below.

Codex (OpenAI). Codex supported drafting and revision of selected passages, improving clarity, organising sections, and editing LaTeX. It also assisted with the presentation of tables, figures, and diagrams, the identification of inconsistencies, and document-formatting checks. Suggested text and changes were reviewed and revised before inclusion in the thesis. Reported numerical values were checked against the retained project results.

Generative AI assistance did not replace the author's responsibility for the research. The author directed the work, assessed suggested changes, and approved the material included in the thesis. The research direction was informed by academic supervision and the maritime context and domain guidance provided by Akquinet/CHESSCON. Generated text was not treated as a scholarly reference or as independent evidence supporting a research claim.

The author takes full responsibility for the research decisions, analyses, reported results, interpretations, conclusions, and the final text, tables, figures, and technical material presented in this thesis.

Terms and abbreviations

Table 1 defines the terms and identifiers used to distinguish scenario information, simulator settings and evaluation evidence.

Table 1: Terms and abbreviations used in the thesis.

Term	Meaning in this thesis
Baseline workflow	Direct human-operated interpretation, configuration review and verification without the workflow-support package. The approved questions also call it the manual baseline.
Controlled information record	One statement in the workflow's controlled information model, with its meaning, source and intended use.
Controlled information model	The selected scenario, simulator-context and evaluation-evidence records organised through stable identifiers and explicit relationships. It is distinct from the executable CHESSCON model.
CHESSCON	The commercial discrete-event simulation product used as the external simulator. The product name is not expanded as an abbreviation in its documentation.
CHK	Check identifier. For example, `CHK003` names one framework check.
Configuration	The values, objects and relationships that tell the simulator which model to execute.
Consistency check	A defined test of whether related information records, settings or results agree.
CSV	Comma-separated values, a text format used for tables such as trace links and evidence records.
DSR	Design Science Research, the methodology used to design and evaluate the proposed workflow.
Evidence catalogue	A table that lists available evidence items, their purpose and their location.
Evidence-processing tool	The separate Python reader used after execution for both workflows; not part of the proposed workflow or its support package.
EV	Evidence-catalogue entry identifier. For example, `EV001` names the first catalogue entry.
FL-1	CHESSCON identifier for the battery-electric forklift equipment type used in the case.
FL-002	CHESSCON name of one individual forklift. Its retained row in the reduced-fleet table was excluded by status-based selection.
Human-operated boundary	The point where the workflow-support package provides instructions and a person performs the corresponding action in CHESSCON.
Indexed result	A table record selected by status as part of a set whose total count agrees with the companion index's live-count field.
Job	One completed movement task recorded by the simulator.

Table 1 (continued)

Term	Meaning in this thesis
JSON	JavaScript Object Notation, used for acceptance records, not the unimplemented importer.
kWh	Kilowatt-hours, the unit used for battery energy capacity.
Managed warning	A known, bounded uncertainty whose effect on interpretation is stated without treating the check as a failure.
MAP	Mapping-rule identifier. For example, `MAP006` names the fleet-deployment mapping.
Mapping rule	A documented responsibility connecting controlled information records to simulator actions or result checks.
Markdown	A plain-text format used for readable instructions and reports.
Provenance	Information about where an item came from, when it was produced and how its identity can be checked.
Proposed workflow	The human-operated sequence for organising scenario information, locating settings, checking configuration, processing results, preserving evidence and handling changes.
QC-1	CHESSCON identifier for the ship-side transfer object in the accepted reference runs.
QC-03	Unexpected identifier for a ship-side transfer object in two rejected starts; not the accepted `QC-1` object.
Reference scenario	The unchanged case used for the two main trials: four electric forklifts, 2,000 jobs, 72 simulated hours and one run.
Scenario	A written statement of the situation and behaviour that a simulator is expected to represent.
Semantic correctness	The extent to which a simulator setting preserves the intended meaning of the scenario information.
SF	Retained identifier prefix meaning scenario fact. The final set also uses it for simulator-context and evaluation-evidence records.
SHA-256	Secure Hash Algorithm 256-bit, used here to create a digital fingerprint for a file or controlled file set.
.TER	CHESSCON main project-file extension.
TIC4.0	Terminal Industry Committee 4.0, used only as a source of shared industry vocabulary; formal conformance is not claimed.
Trace link	A recorded connection from a controlled information record to its source, mapping, simulator responsibility, evidence and check.
Validation	Assessment of whether the model and evidence are suitable for the intended study purpose.
Verification	Checking whether an implemented configuration and its outputs satisfy stated rules.
Workflow-support package	The YAML, CSV and Markdown records, mappings, checks, instructions and evidence catalogue used by the proposed workflow.
WR	Workflow Requirement identifier. For example, `WR1` identifies the first requirement of the proposed workflow.
YAML	A human-readable structured-data format used for the controlled information model and mapping rules.

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Chapter 1

Introduction

A simulation allows a proposed situation to be examined in software before decisions are made about a physical system. Its starting point is usually a *scenario*: a description of the equipment, activities, quantities and operating conditions that the model should represent. A simulation model is the software representation of that situation. A *configuration* is the collection of objects, values and relationships that determines how the model behaves when it runs. Turning a scenario into a configuration therefore involves more than entering numbers. The intended meaning of the information must be preserved as it moves from descriptions written for people into structures defined by the simulation product.

This translation can be difficult even when the required information appears simple. A statement about equipment availability may concern a type of equipment, the number of individual units, their deployment or the period during which they are available. The modeller, the person preparing the simulation, must decide which interpretation applies and locate the corresponding settings. A value entered in a valid field can still represent the wrong requirement. Recent research on information exchange in engineering identifies this combination of meaning, technical representation and working practice as an interoperability problem: the difficulty of using information consistently across different systems [35].

The problem continues after the initial configuration. A reviewer may need to explain why a setting was chosen, check whether related settings agree or identify the effect of an updated requirement. These tasks depend on relationships that are not always visible in the simulator itself. *Traceability* means that such relationships are explicitly recorded and can be followed, for example from a scenario statement to its corresponding setting. Industrial traceability research shows that having requirements and tests available does not necessarily mean that the links between them exist [17]. This distinction matters equally when a simulation result is available but the decisions that produced its configuration are difficult to recover.

Simulation software and the information workflow around it have different responsibilities. The simulator executes its configured model. The surrounding workflow establishes what should be represented, how the information relates to that configuration and what evidence supports the result. A *workflow* is an organised sequence of activities and responsibilities. This thesis examines whether making those information responsibilities explicit provides a useful basis for reviewing a configuration and handling later changes. The research is concerned with software work, not with improving the physical operation represented by the model.

CHESSCON, the commercial simulation product used in this study, provides the external setting in which the workflow is examined [4]. Its product name is retained rather than treated as an acronym. The proposed workflow does not replace CHESSCON or operate it automatically. Instead, it provides controlled information records, mappings and checking guidance that a person uses alongside the simulator. This choice reflects evidence that traceability practices remain substantially human-operated and must fit the work in which they are used [54]. It also creates a clear evaluation question: what is demonstrated by organising the information, and what still requires direct evidence about the resulting work?

CHESSCON was selected because the industry collaboration provided access to the software and practical guidance. The evaluation used a reference model that had been configured before the workflow trials. This setting supported a study of configuration review, verification and evidence handling. No comparative simulator-selection exercise was performed, so the choice does not establish a preference over alternative simulators.

1.1 Research problem

The research problem is the weakly recorded connection between a written scenario and its simulator-specific representation. A direct workflow relies on a modeller to interpret the available information, identify the relevant settings and verify the result. Those activities are necessary and can be carried out competently. The difficulty is that the reasoning may remain distributed across documents, product guidance and personal knowledge rather than being maintained as a connected record.

Three kinds of information must remain distinguishable. The scenario states the intended situation. A mapping records how a scenario concept corresponds to a simulator object, field or action. Evidence records what was actually inspected or produced. Treating these as interchangeable can hide an important gap: documenting the intended value does not establish that it was configured, and observing a result does not independently confirm every property of the model. Model-transformation research identifies tracing and reusable transformation mechanisms as important capabilities, while also showing that their effects depend on context [23].

Changes make these relationships more important. A revised quantity can affect several configuration locations and the expected result, whereas a structural change can alter which equipment records should be active. *Consistency* means that these related representations agree according to stated rules. Studies of industrial model-based development describe the difficulty of maintaining such agreement when information is distributed across models and tools [29]. The challenge addressed here is to expose the relevant relationships without assuming that an executable connection to the simulator is available.

The problem also concerns the identity of evidence. *Provenance* is the recorded origin and history of an item, including the configuration and process that produced it. A numerical result is useful only if it can be associated with the intended model and interpreted using the correct record-selection rules. Research on reproducibility distinguishes the presence of a result from the information needed to reproduce or

inspect its production [21]. The thesis therefore treats evidence preservation as part of the information problem rather than as a final administrative task.

1.2 Research questions

RQ1: How can existing research be used to define the manual baseline for converting a scenario into a simulator configuration?

The first Research Question (RQ) establishes a defensible comparison workflow. A comparison would be difficult to interpret if its baseline omitted necessary work or were defined only to favour the proposed solution. Existing research provides evidence about interpretation, mapping, traceability, consistency, verification and provenance, but it does not supply one complete CHESSCON procedure for this study. This question was chosen to make the synthesis explicit: the baseline must follow from the responsibilities identified in the literature. Its answer is therefore a reasoned workflow definition, not a numerical claim that all possible manual-workflow problems occurred in the case.

RQ2: How should the proposed software workflow be designed to support this conversion while maintaining traceability and managing later scenario changes?

The second question concerns the design of the contribution. Identifying difficulties in a direct workflow does not explain how information should be organised to address them. This question requires a concrete account of the records, relationships, checks and responsibilities that the proposed workflow introduces. It was chosen to connect the literature findings to an implemented, inspectable approach rather than to a general recommendation to improve documentation. The design must also explain how a person uses the information with CHESSCON and how later changes are related to affected settings, while keeping the boundary between documented guidance and executable software clear.

RQ3: How does the proposed workflow compare with the manual baseline under the same scenario and evaluation conditions?

The third question examines the proposed workflow against the baseline under common conditions. A structured package may look complete without demonstrating a benefit in use. The question was chosen to separate the properties visible in the package from observations obtained during the comparison. The evaluation examines modelling effort, internal trace completeness, consistency and change handling, supported by configuration and result checks. These are the evaluation dimensions used to investigate the question, not additional research questions. The answer must distinguish recorded differences from causal improvements and recognise which activities the available evidence does not measure.

1.3 Aim and objectives

The aim is to develop and evaluate a traceable, human-operated workflow for supporting scenario-to-simulator configuration. The workflow should keep selected scenario information, simulator mappings, checking rules and later changes connected

so that configuration decisions and their evidence can be reviewed. Its practical implementation is a *workflow-support package*: a collection of structured records and readable instructions used by a person, rather than an automatic importer.

Table 1.1 connects the objectives to the research questions. The objectives describe the work needed to answer the questions; they do not assume that the proposed workflow will outperform the baseline.

Table 1.1: Research objectives and their connection to the questions.

Objective	Intended outcome	Question
1	Synthesise relevant research into a credible baseline workflow.	RQ1
2	Identify the information and checking responsibilities that the proposed workflow must support.	RQ1, RQ2
3	Implement those responsibilities through connected records, mappings, traces and guidance.	RQ2
4	Compare both workflows using the same reference model and execution acceptance conditions.	RQ3
5	Examine how workload, fleet and distance changes affect information and verification responsibilities.	RQ3
6	Explain the findings at the level supported by the preserved evidence.	RQ1–RQ3

The objectives form a progression from definition to design and evaluation. Establishing the baseline provides the comparison; implementing the proposed workflow makes its support inspectable; and the controlled evaluation tests selected aspects of its use. Design Science Research, a methodology that develops and evaluates a practical solution, provides the overall research structure [22].

1.4 Thesis contribution

The thesis contributes one proposed scenario-to-simulator workflow. Its contribution is the organised relationship between selected scenario information and the simulator responsibilities needed to represent, check and revise it. The implementation makes those relationships explicit through controlled records, mappings, trace links, consistency checks, configuration guidance, change-impact records and an evidence catalogue. These are supporting parts of the workflow, not separate claims of independent software systems.

The design separates information that states what is required from information that explains where it belongs in CHESSCON. It also separates the record of a required check from evidence that the check was performed. The published examples illustrate these responsibilities and relationships without assuming an unseen transformation service. Inspection of the complete implementation requires access to the private registers under the arrangements stated in the Data Availability Statement. A JSON importer was planned but not implemented. The evaluated contribution is the human-readable package, and the thesis evaluates only what was implemented.

The contribution includes a bounded empirical assessment. Both workflows were examined around an already configured reference model, with preserved results and selected scenario changes. The assessment establishes what the workflow records and which execution conditions were satisfied. It does not establish complete model

construction from an empty project, general productivity improvement or better simulator performance. These boundaries define the contribution precisely rather than replacing its practical value with unsupported claims.

1.5 Ethical considerations

The study uses commercial simulation software and project information whose access and redistribution are controlled by their owners. The thesis presents derived results, explanations and permitted CHESSCON screenshots. The original licensed projects and restricted raw files are not distributed with the thesis. Permission to reproduce screenshots is stated in their captions; it does not imply permission to publish the underlying models or all related organisational information.

Responsible reporting requires the distinction between observed evidence and interpretation to remain visible. The study does not evaluate employees or use simulation output to make claims about individual work performance. The results are reported as evidence about a bounded software workflow. Rejected setup events, incomplete timing and the limits of the accepted-run checks are retained where they affect interpretation. This approach is consistent with the attention to affected people and context advocated in socially responsible design science [47].

The simulation results and supporting evidence were presented to the university supervisor during supervision. This supervision is not described as formal ethical approval or as independent validation of the numerical results. Scientific responsibility remains with the reporting: sources must be credited, evidence must not be altered to strengthen a conclusion, and limitations on access must be clear to readers.

1.6 Study scope and thesis organisation

The study concerns one defined scenario and one external simulator. The main comparison starts with independent setups derived from an already configured model. It examines preparation for review, configuration verification, execution and evidence handling. Later changes provide additional evidence about dependencies and result interpretation. The evaluation does not cover all activities required to construct a new simulation model, and it uses descriptive observations rather than statistical inference from repeated independent trials.

Chapter 2 develops the literature-based baseline and design requirements. Chapter 3 explains the research method. Chapter 4 defines the scenario and proposed workflow, while Chapter 5 specifies the measures and acceptance conditions. Chapter 6 reports the findings for the research questions. Chapters 7 and 8 interpret the findings and conclude the study. Appendix A provides supplementary CHESSCON screenshots and result details.

Chapter 2

Background and Related Work

A simulation configuration is the result of several information decisions. Someone must establish what a statement means, decide how the simulator represents it, check the resulting settings and retain evidence of the outcome. Research on these decisions is spread across digital representations, model transformation, traceability, consistency and reproducibility. This chapter brings those areas together to define the baseline workflow and the responsibilities of the proposed workflow.

2.1 Focused literature review

2.1.1 Purpose and review approach

The review is a focused, thematic synthesis. It explains the information responsibilities of a human-operated scenario-to-simulator workflow. It does not estimate how prevalent these problems are across the research field. A thematic synthesis groups findings by the problem they address and then examines their implications for a common design problem. Here, the themes follow the information from its initial meaning to its recorded simulation evidence.

The focused approach was selected because the review served as design input within the Design Science Research study. Its intended output was a justified set of workflow responsibilities and requirements. This approach concentrated the synthesis on findings applicable to those design decisions. A systematic literature review could also support the design, but would require a documented, reproducible search and selection protocol to establish a systematically assembled evidence base. The chosen review scope prioritised targeted design support, while accepting incomplete literature coverage and selection bias as limitations.

The final bibliography contains 73 references: 70 research publications and three authoritative sources. The original focused corpus comprised 67 research papers. Bengtsson et al., Peffers et al. and Robinson were added during revision as simulation-specific or methodological background [7, 50, 53]. ISO 23247-1:2021 [24], the TIC4.0 standards and definitions portal [63], and CHESSCON product information [4] provide standards context, industry terminology and product context. These three sources are separate from the peer-reviewed research corpus. The bibliography-generation record contains a fixed list of paper Digital Object Identifiers (DOIs). A DOI is a persistent identifier for a publication. The focused literature review examined studies relevant to interpreting scenario information, mapping it to simulator settings, maintaining traceability, checking consistency and handling

changes. These studies informed the responsibilities included in the baseline workflow and the design of the proposed workflow. The review provides a targeted synthesis of these topics rather than an exhaustive account of all published research. The historical database-search and screening decisions were not recorded in sufficient detail to reconstruct an exhaustive selection process. The thesis therefore makes no systematic-review or complete-coverage claim. Appendix B identifies the search strings and provides reconstructed search queries mapped to the cited publications.

For the present synthesis, the central publications were checked against their published descriptions of the research problem, method, evidence and findings. Full published text was used where accessible; where only the publisher's abstract was available, the account is limited to what that abstract supports. No unreported dataset, prototype or experimental procedure is inferred from a title. Methodological studies inform the research design, industrial studies inform practical difficulties, and application studies illustrate the conditions under which stronger integration claims are possible.

Table 2.1 distinguishes these roles. It also explains why a paper about an industrial application can be relevant without making its performance result transferable to this thesis.

Table 2.1: Scope and use of the retained literature.

Evidence type	Use in the review
Systematic and mapping reviews	Establish concepts, recurring concerns and the range of existing approaches; not frequencies for this case.
Industrial and practitioner studies	Explain difficulties encountered when maintaining links, models and configurations.
Implemented research tools	Identify the input structures, interfaces and evaluation evidence needed for executable integration.
Design and methodological papers	Support explicit responsibilities, evaluation boundaries and transparent reporting.
Application studies	Contrast the present workflow with live monitoring, optimisation and connected industrial systems.
Official product and standards sources	Establish CHESSCON's role and the meaning of relevant industry terminology.

The review does not treat all 73 references as equally important to the proposed workflow. Table 2.2 presents ten studies central to its design and evaluation. This is a purposive selection based on direct relevance to the information problem, not a numerical ranking or an assessment of overall scientific quality. The table connects each study's method and evidence to its specific use here; the following thematic discussion explains what is adopted and what is not.

Table 2.2: Ten studies central to the workflow design and evaluation.

Study	Method, evidence and prototype	Specific use in this thesis
Liepert et al. [35]	Review of interoperability in early engineering; synthesis rather than an evaluated integration prototype.	Separate differences in meaning from differences in data structure and organisational responsibility.

Table 2.2 (continued)

Study	Method, evidence and prototype	Specific use in this thesis
Fucci et al. [17]	Industry–academia study in telecommunications; missing requirement-to-test links obstructed the intended analysis.	Do not assume that related documents already provide a usable trace.
Höppner and Tichy [23]	Online survey with 113 responses concerning model-transformation languages and their benefits.	Make mappings and traces explicit, without assuming that formalisation guarantees reduced effort.
Jongeling et al. [29]	Empirical investigation of consistency management in 13 industrial model-based settings.	Treat consistency as a responsibility across related records, not one final screen check.
Ruiz et al. [54]	Survey and interviews concerning industrial traceability barriers; 61 unique participants.	Give links a defined purpose and acknowledge the work required to maintain them.
Maro et al. [42]	Design-science development of the TracIMo traceability-introduction methodology in a financial-sector development setting.	Introduce trace information around concrete tasks and goals rather than record every possible relationship.
Lugaresi and Matta [40]	Implemented model generation from object-centred event logs; test cases and a manufacturing application.	Explain why automation requires structured source evidence unlike a written scenario.
Mkaouar et al. [44]	Benchmark implementations using four incremental transformation tools and an industrial engineering model.	Treat change propagation as a relationship problem and distinguish tool capability from measured benefit.
Tröls et al. [64]	Implemented cross-tool consistency support for concurrent engineering changes, with empirical feasibility assessment.	Define agreements across representations; do not describe manual checks as a live checking service.
Hevner et al. [22]	Conceptual account of transparency in design science, covering design and evaluation decisions.	Separate what was designed, what was implemented and what was actually evaluated.

2.1.2 Digital representations and the meaning of integration

A digital representation is a software description of a system or situation. It may support analysis without maintaining a live connection to physical equipment. Semeraro et al. [56] review the digital-twin literature by examining components, features and their relationships. Their work is relevant because it prevents the digital model from being treated as the complete integration system: data, connections and intended services also matter. Phua et al. [51] make this distinction concrete in metal additive manufacturing by describing a hierarchy with increasing modelling, sensing and control capability. This thesis adopts the need to state the level of integration; it does not adopt that industry's process hierarchy as its own classification.

Liu et al. [37] organise published digital-twin engineering work around physical entities, virtual models, data and applications. Wooley's synthesis of systematic reviews similarly finds substantial variation in definitions and recurring problems

with data quality and integration [69]. Together, these studies support a precise description of the present implementation. It is a workflow around a simulator and a human-readable support package, not a continuously connected digital twin of a physical site.

Application studies show why that boundary matters. The Simplexity testbed combines physical rovers with several modelling technologies to investigate uncertainty [67]. Calvo-Bascones et al. [9] connect digital-twin behaviours for anomaly detection in a generator cooling system. Their evidence concerns physical behaviour and diagnostic models. The present thesis has neither that physical testbed nor a comparable diagnostic experiment, so their operational findings cannot validate its simulator results. Their useful contribution is the example of connecting a stated purpose to a specific evaluation setting.

Other studies provide further contrasts. Jiang et al. [28] describe a device-oriented framework with data, models and services in an equipment-monitoring application. Khalifa and Marzouk [32] present a connected building-energy platform with implemented interfaces. Tao et al. [62] examine modelling and model services as a distinct part of digital-twin development. These works help distinguish a model, a service and the interface joining them. In this thesis, a configuration guide is not such an executable interface: a person reads it and operates CHESSCON.

Bertoni and Bertoni's review of product–service systems identifies a need for stronger evaluation alongside proposed frameworks [8]. The product-design review in [38] likewise places digital-twin uses within particular development activities rather than assuming one universal benefit. This motivates evaluating a clearly bounded task. It does not justify calling the current comparison a complete test of all activities needed to construct a new simulator model.

2.1.3 Meaning, terminology and shared information

Semantic interoperability is the preservation of meaning across representations. A source statement can be copied accurately while still being misunderstood if its unit, context or intended object changes. Liepert et al. [35] identify manual identification, conversion and integration of information as continuing difficulties in early engineering. They distinguish semantic issues from technical and organisational ones. This thesis uses that distinction to separate the meaning of a scenario fact, its simulator representation and the person responsible for checking the relationship.

Acharya et al. [2] survey interoperability challenges at several levels, including data syntax, meaning and organisational interaction. *Syntax* is the form in which information is written. Agreement on syntax does not establish agreement on meaning. A correctly formatted number can still represent the wrong duration or object. This is why a controlled record in the proposed workflow includes meaning and context, with a value and unit where applicable, rather than an unexplained value alone.

An *ontology* formally describes concepts and relationships within a domain. Karabulut et al. [31] review ontology use in digital twins and distinguish its roles across application areas and structural layers. Ontology-based data modelling in [41] addresses relationships, coupling and evolution of digital-twin information. The knowledge-based approach in [66] connects model-based engineering concepts through a satellite-design example. These approaches motivate separating shared meaning

from product-specific fields. However, the proposed workflow does not implement a formal ontology or an automated semantic reasoner.

Related approaches also depend on their intended use. Semeraro et al. [57] investigate reusable, data-driven modelling patterns in aluminium die casting. The cognitive digital-twin review in [71] examines knowledge representation and reasoning, while the virtual manufacturing example in [73] connects monitoring and prediction to a production system. Their relevance is the explicit organisation of relationships, not evidence that the present YAML records perform learning, prediction or reasoning. YAML is a human-readable structured text format; its use here is explained in Chapter 4.

Industry vocabulary provides a lighter form of shared terminology. The *Terminal Industry Committee 4.0 (TIC4.0)* publishes terms and definitions for equipment and activities in cargo-handling systems [63]. It is included because scenario descriptions and simulator labels can refer to the same equipment in different ways. The thesis uses this vocabulary as context, not as proof that the entire scenario or workflow conforms to a TIC4.0 data standard.

The *International Organization for Standardization (ISO)* publishes international standards. ISO 23247-1 gives an overview and general principles for manufacturing digital-twin frameworks [24]. It is relevant as a reference for discussing entities and responsibilities. The study neither performs a formal conformity assessment nor claims that its human-operated package implements the standard. These two sources clarify terms and scope; they do not supply the study's acceptance results.

2.1.4 Mapping and model transformation

A *mapping* records how information in one representation relates to another. A *model transformation* applies defined relationships to produce or update a target model. These concepts are closely related, but a written mapping is not automatically executable. Höppner and Tichy [23] investigate model-transformation language use through a practitioner survey. Their findings connect tracing and reuse with perceived benefits, while also showing that benefits depend on context. This supports recording mapping decisions explicitly, but does not support assuming an effort reduction merely because mappings exist.

Lugaresi and Matta [39] generate manufacturing models from event records while controlling model detail. Their later work [40] addresses complex material flows through object-centred logs and graph-model completion. Both are valuable because they show what automated generation actually requires: interpretable event data, defined modelling assumptions and a generation procedure. A written scenario and a commercial simulator interface are different inputs and boundaries. Consequently, the proposed workflow retains human interpretation and configuration rather than presenting a document package as an automatic generator.

Jia et al. [26] describe a method for composing models at different scales and for different scenarios. Their subsequent implementation uses connected models for manufacturing tasks such as tool wear and temperature [27]. The distinction between the method and its application is useful here. The intended workflow covers scenario conversion, but the empirical comparison covers checking and executing an

already configured model. A design objective must not be mistaken for an activity demonstrated by the evaluation.

Mkaouar et al. [44] benchmark four tools for incremental model transformation, meaning updates to affected model parts rather than rebuilding everything. Their industrial model and tool implementations expose practical differences in applying these techniques. This informs change-impact recording in the thesis: an update should identify affected relationships. It does not justify importing their tool timings or assuming automatic propagation through CHESSCON.

The aircraft-manufacturing study in [70] uses a digital thread, a connected path of engineering information, to relate data and models in drilling and riveting work. Its relevance is the explicit relationship across stages; the present thesis does not implement its production-system connections.

Broader reference architectures illustrate how responsibilities can be separated without transferring their whole implementation. The general-purpose digital-twin architecture in [1] distinguishes data acquisition, modelling and higher-level services through a semiconductor-production case. The predictive-maintenance work in [65] uses domain analysis and architectural views in several applications. Here, the adopted principle is the separation of information responsibilities. The actual implementation remains a workflow-support package, not a general-purpose platform or maintenance system.

2.1.5 Traceability and the work needed to maintain it

Traceability is the ability to follow a recorded relationship between related pieces of information. Fucci et al. [17] provide a particularly important industrial example: missing requirement-to-test relationships obstructed the analysis planned in an industry–academia collaboration. The lesson is not that every manual process fails. It is that the existence of requirements and tests does not establish a usable link between them. The equivalent concern here is whether the source, simulator setting and verification evidence are explicitly connected.

Ruiz et al. [54] investigate why industrial traceability remains difficult through a survey and interviews. Their findings show that perceived value, effort and working context matter. Maro et al. [42] address introduction of traceability through an incremental, goal-oriented methodology in a financial-sector setting. These studies inform the decision to record links with a concrete configuration or verification purpose. They also prevent a one-sided argument: maintaining the proposed records is work, and the thesis has not measured that long-term maintenance cost.

Trace creation and trace recovery are different activities. The machine-learning review in [5] examines methods and datasets used to infer links; the natural-language-processing review in [49] examines the use of written text for trace recovery. DSSLink [72] uses semi-supervised learning to recover issue-to-commit links, while TCTracer [68] combines static and execution information for test-to-code links. These are implemented or reviewed recovery techniques, not evidence that this thesis automatically reconstructs relationships. The study measures only links actually present in the retained workflow records.

METRA [33] uses model execution information to support requirement-to-process-model relationships. Traceability-graph research in [11]

shows how established links can support questions about software contributions. An eye-tracking study [3] explores another possible source of trace information through practitioner interviews and a project-based investigation. These studies extend the possible uses and sources of links, but none makes a link count equivalent to verified understanding or improved productivity.

The review of pre-requirements traceability in [48] examines links to information created before formal requirements, including discussions and legacy material. It motivates an important distinction in this thesis. Completeness within a controlled register can be counted, but completeness of the original source capture requires an independently established source inventory. The 30-record register supplies the first denominator, not the second. Internal trace completeness therefore has a narrower meaning than complete coverage of every original requirement.

2.1.6 Consistency, configuration and change

Consistency means that related representations agree according to a stated rule. Jongeling et al. [29] study consistency management in 13 industrial settings and show that keeping related models aligned is an ongoing practical concern. This finding is relevant because simulator values may be distributed across equipment types, deployment entries and workload settings. A single correct field does not establish agreement among all connected fields.

Tröls et al. [64] implement consistency support across engineering tools during concurrent changes. Linington et al. [36] examine the role of specialised languages in expressing enterprise consistency constraints as systems evolve. These contributions support explicit rules and relationships. Their execution mechanisms are not adopted here: the proposed package records checks that a person applies, rather than running a live service that detects every change.

Configuration research also concerns choices among software settings and variants. Colla-Config [6] supports collaborative resolution of stakeholder configuration preferences. The software-product-line mapping study in [16] examines configuration management across families of related products. These are broader configuration problems, not direct demonstrations of wrong-project selection in CHESSCON. The present study derives its specific project-identity control from its simulator boundary and retained setup evidence.

ETune [10] searches configuration choices under changing workloads, and EASIER [12] investigates architecture refactoring through optimisation. They are useful contrasts because their objective is to find better system configurations. This thesis instead asks whether the intended configuration and its supporting information remain clear. It does not search for the fastest simulation, the best equipment allocation or an optimised terminal layout.

Change handling therefore has two parts in the present synthesis. The affected information must first be identified, and its revised representation must then be checked. A list of dependent records can support the first part. Only a preserved execution and its revised acceptance checks can establish the second where simulation is required. This distinction follows the difference between recording a transformation relationship and executing a transformation.

2.1.7 Verification, provenance and reproducibility

Verification examines whether a model or configuration follows stated rules. *Validation* examines whether it is suitable for its intended purpose. Sinisi et al. [59] investigate verification and interoperability in an open simulation environment through an implemented state-management capability and model benchmarks. The robot-simulation study in [15] evaluates particular navigation and sensing behaviours. Their evidence illustrates why an acceptance claim must name the behaviour or property checked. Passing selected configuration checks is not equivalent to validating every physical feature of a scenario.

The simulation-credibility work in [58] considers the strength and conflict of different forms of evidence. This thesis adopts the need to distinguish evidence roles, but does not calculate that paper's credibility measure. A screenshot can explain an interface, an input record can establish a configured value, and a retained result table can support a numerical total. They answer different questions and are not interchangeable.

Provenance records where information came from and how it was produced. Dhruv et al. [14] discuss the execution environment needed to reproduce computational work. Samuel and König-Ries [55] develop semantic provenance support spanning computational and other research activities. These works motivate retaining both software output and the steps needed to interpret it. A table of final totals alone cannot explain which project or record-selection rule produced those totals.

Reproducibility studies show the cost of missing context. González-Barahona et al. [20] examine reproducibility assessment in software-repository research. Pimentel et al. [52] investigate computational notebooks and the effects of execution and dependency information. Attempts to reproduce Android malware detectors in [13] similarly expose the work caused by incomplete experimental detail. The present use of these studies is limited to evidence preservation and reporting; their success rates or effort observations do not measure this workflow.

Sun et al. [61] investigate how software-package origins can be identified, showing that identity itself can require analysis. The reproducibility-debt synthesis in [21] brings together causes, consequences and mitigation strategies for missing reproduction information. Research on reproducing question-and-answer code examples [46] provides another case in which readable text does not guarantee sufficient execution context. These studies support preserving original evidence identity and explaining the reader, rather than treating a derived chart as self-sufficient proof.

Restricted software changes what can be shared, but not what a claim means. The thesis presents derived results and permitted screenshots. Original licensed projects and protected evidence remain subject to access restrictions. The material was presented during university supervision; this is not a claim that an independent reproduction study has been completed.

2.1.8 Design-science and application perspectives

Design Science Research (DSR) develops and evaluates an artefact intended to address a practical problem. Hevner et al. [22] argue for transparency about the problem, design, construction, evaluation and contribution. Storey et al. [60] distinguish

reliability of the artefact, methods, measures and resulting knowledge. These contributions inform the method in Chapter 3: the existence of a package, the correctness of its links and the quality of a timing comparison require different evidence.

Monson [47] examines social responsibility in design-science problem framing. This supports stating who can inspect or distribute the evidence and avoiding inflated claims about benefit. The thesis applies these concerns through the ethical considerations in Section 1.5; it does not claim a separate social-impact evaluation.

The application literature establishes context without defining the contribution as a port-optimisation project. Giering and Dyck [19] propose a maritime digital-twin concept spanning shipbuilding and shipping. The Port of Hamburg platform study [30] examines information exchange among different participants. Min [43] uses content analysis to organise smart-port components and development considerations. Their scale includes organisations and operational services beyond this thesis. The relevant connection is the need to relate information across tools and responsibilities, not evidence of improved physical operations here.

Ivanov [25] treats a digital supply-chain twin as a socio-technical system, linking technology with people and organisational factors. Mohanraj and Vaishnavi [45] review data-enabling technologies and continuing integration issues. The systems-of-digital-twins review in [18] examines composition of representations for specific purposes. These perspectives support a visible human boundary and a bounded claim. The green-manufacturing application in [34], for example, evaluates an environmental purpose that this thesis has not measured. It cannot be used to infer energy savings merely because electric equipment appears in the scenario.

CHESSCON is the commercial simulation product used for the study [4]. It supplies the environment in which the configuration is inspected and executed. Its product role is separate from the proposed workflow, just as the industrial systems in the reviewed applications are separate from their research methods.

2.1.9 Simulation-specific conceptual modelling and input data

Two foundational simulation publications address responsibilities that are especially close to this study. Robinson [53] describes conceptual modelling as making a simulation's purpose, scope, assumptions and simplifications explicit before implementation. Bengtsson et al. [7] describe identifying and collecting input data, processing it and structuring it for discrete-event simulation. These works help distinguish scenario meaning and input management from simulator-specific configuration. They were added during this revision as focused background, not as part of the original 67-paper corpus or as evidence from the thesis evaluation.

2.2 Defining the baseline workflow through the literature

2.2.1 From findings to responsibilities

The review supports a baseline in which a person directly translates and checks scenario information using the available sources and simulator guidance. It does not supply one published CHESSCON workflow to copy. The baseline is the thesis's synthesis of responsibilities that recur in the reviewed research. Table 2.3 makes the derivation explicit by connecting each responsibility to the studies that most directly support it.

Table 2.3: Literature responsibilities used to derive the baseline.

Responsibility	Main evidence	Consequence for the baseline
Interpret meaning	Interoperability review [35]	Establish the value, unit, context and intended object before choosing a simulator field.
Map representations	Transformation survey [23]; model generation [40]	Make product-specific mapping decisions; do not assume an executable transformation.
Retain available links	Industrial traceability studies [17, 54]	Preserve the relationships genuinely available during the work.
Check agreement	Industrial consistency study [29]	Compare connected settings, not only isolated values.
Verify stated conditions	Simulation verification [15, 59]	Define what the configuration and accepted output must satisfy.
Preserve provenance	Computational provenance [14, 55]	Keep the relationship between project state, result and interpretation.

These responsibilities do not form six independent tests. Interpretation influences mapping; mapping identifies which fields must agree; verification requires evidence that belongs to the intended configuration. Traceability and consistency therefore cross the whole workflow. For example, a source reference is useful when interpreting a value and again when reviewing why that value was configured. Separating these responsibilities in the analysis does not imply that a modeller performs them in isolation.

2.2.2 Constructing the baseline workflow

The baseline begins by examining the written scenario and supporting material. Examination identifies the information relevant to the intended simulation, whereas interpretation establishes what that information means. Keeping those stages distinct avoids treating every number in a source document as a simulator input. A reported measurement, an assumption and a required setting can look similar on a page while serving different purposes.

Mapping follows interpretation because the intended meaning determines the appropriate simulator representation. The modeller identifies the relevant object, field and unit, then enters or confirms the setting through the supported interface. The phrase "enters or confirms" is deliberate: the workflow can be applied to new

configuration work or to review of an existing model. The latter is the activity exercised in the controlled comparison.

Configuration verification connects the settings to predefined expectations. It includes checking the selected project, executing the model where required and examining the resulting evidence. Recording the evidence retains the basis of that judgement. These activities are summarised in Table 2.4; each output supplies information needed by the next stage rather than simply marking completion of a screen.

Table 2.4: Stages and outputs of the baseline workflow.

Stage	Work performed	Result of the stage
Examine the scenario	Identify relevant statements in the scenario and supporting documents.	Information requiring interpretation.
Interpret the information	Establish meaning, values, units, context and assumptions.	Interpreted requirements.
Map the information	Locate corresponding simulator objects and settings.	Product-specific mapping decisions.
Configure the simulator	Enter or confirm the intended settings in the selected project.	A configured model for checking.
Verify the configuration	Check settings, run the model and compare output with expected conditions.	An accepted result or a recorded issue.
Record the evidence	Preserve the available sources, decisions, configuration and results.	A basis for reporting and later review.

Figure 2.1 shows the baseline workflow. Its sequence follows information from the source to the recorded result. The linked concerns identify where meaning, links, project identity or active-result selection may require attention; they do not claim that every concern occurred in the study.

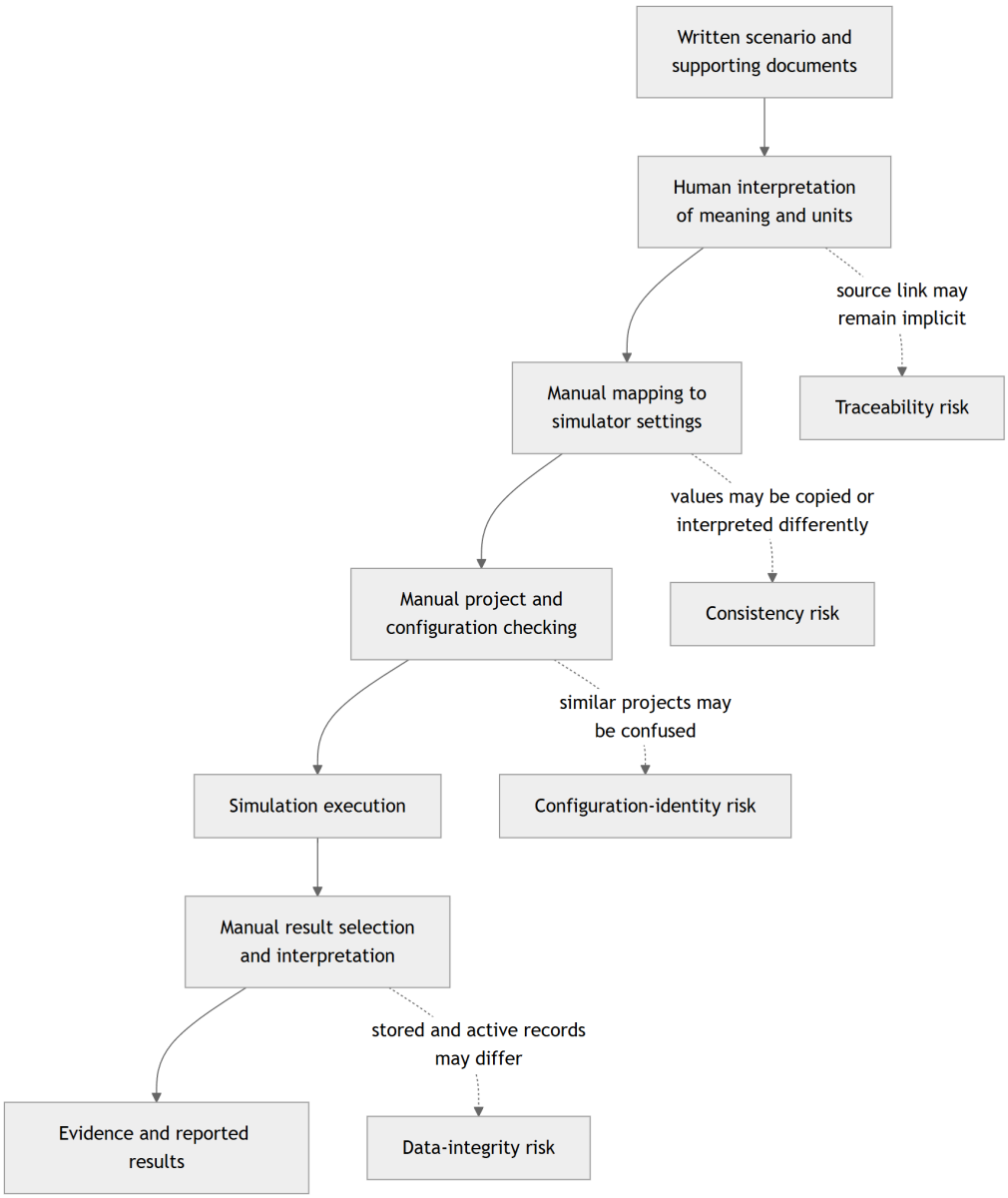


Figure 2.1: Manual workflow for converting a written scenario into a simulator configuration and producing verifiable results.

2.2.3 The resulting baseline workflow

In the baseline workflow, a modeller interprets the written scenario and maps its information to simulator settings. The modeller then configures and verifies the model and preserves the available evidence. This human-operated process does not use the proposed workflow-support package. The modeller can use source documents, simulator guidance and normal product knowledge. The definition does not prohibit notes or checks, and it does not assume that direct configuration is incorrect.

Illustration of the defined baseline. For the existing four-forklift requirement, the modeller interprets the scenario, consults simulator guidance and inspects deployment for four units of type `FL-1`. After execution, the common evaluation procedure

selects active result records and cross-checks their count against the index. The modeller compares count and type with the requirement and retains the evidence. This illustrates the defined baseline without SF or MAP records; it is not a new trial or reconstructed observation log.

The defining feature is where the connections are maintained. In the baseline, the modeller works directly between existing sources and the simulator. The proposed workflow adds a shared organisation for selected information, mappings, traces and checks. This establishes a credible comparison of information support, rather than a comparison between careful work and an intentionally weak method. Evaluator-created logs must consequently remain separate from traceability that was actually supplied to the baseline during its work.

2.3 Research gap and workflow requirements

The reviewed studies offer strong methods for formal modelling, trace recovery, consistency checking and provenance. Their implementations often require structured source data, compatible models or executable interfaces. The problem addressed here is narrower: how to organise the corresponding information responsibilities when a person must carry information across the simulator boundary. The contribution is not a claim that the literature lacks workflows or traceability tools. It is a designed and evaluated combination for this bounded setting.

Table 2.5 translates the synthesis into eight Workflow Requirements (WRs). The identifiers connect the reasoning here to the implementation-and-evidence matrix in Table 6.2. They express intended responsibilities, not eight measured improvements.

Table 2.5: Requirements for the proposed workflow.

Identifier	Required support	Reason
WR1	Give selected information an agreed recorded location.	Make the source of a value explicit.
WR2	Record source, controlled content or value, unit where applicable, and meaning.	Preserve interpretation context.
WR3	Separate information from simulator mappings.	Distinguish meaning from product-specific representation.
WR4	Link information, settings, checks and evidence.	Make relationships available for review.
WR5	Define checks before interpreting output.	Keep expectations independent of observed results.
WR6	Record affected information for later changes.	Expose the intended scope of an update.
WR7	Preserve project and evidence identity.	Distinguish intended, historical and active records.
WR8	Keep human operation explicit.	Describe the actual implementation boundary.

The requirements connect the literature-derived baseline to the proposed workflow. Chapter 3 explains how that workflow was studied, and Chapter 4 shows how the responsibilities were implemented without changing CHESSCON itself.

Chapter 3

Research Method

This chapter explains how the research was organised. It defines the research design, the part of the process being studied, the controlled comparison, the evidence used and the connection between the method and the three Research Questions.

3.1 Overall study design

This thesis adopts Design Science Research (DSR) as its overall research methodology. DSR is suitable because the study addresses a practical software-engineering problem by designing and evaluating a proposed scenario-to-simulator workflow [22, 60]. The methodology is supported by a focused literature review for defining the baseline workflow, a controlled single-case comparison for evaluating the two workflows, and descriptive evidence analysis for interpreting the findings without making statistical claims.

A literature review alone could identify information responsibilities, but could not show how they were represented in an implemented workflow. A practitioner survey could examine perceived difficulties or usefulness, but would not directly verify the contents of this package or the accepted simulation outputs. The selected combination therefore connects literature-based reasoning, development of an inspectable artefact and a bounded comparison. Descriptive analysis fits the evidence available without treating the single accepted trial per workflow as a population-level estimate.

A single case means that one bounded scenario and one simulator are used as the evaluation setting. The comparison remains controlled because the baseline and proposed workflows use separate project setups created from the same verified model and are assessed using the same conditions.

In DSR, a designed solution is referred to as an *artefact*. The artefact developed in this thesis is the proposed scenario-to-simulator workflow. The workflow is implemented through controlled information records, mappings, trace links, predefined checks, configuration guidance, change-impact records and evidence rules. CHESSCON remains an external simulator, and the evidence-processing reader remains a separate evaluation tool.

The research contains three activities. The literature defines a credible baseline and informs the design responsibilities. The proposed workflow implements those responsibilities. The controlled comparison then examines its use alongside the baseline.

Figure 3.1 shows their relationship, including the two separate evaluation branches and their common comparison conditions.

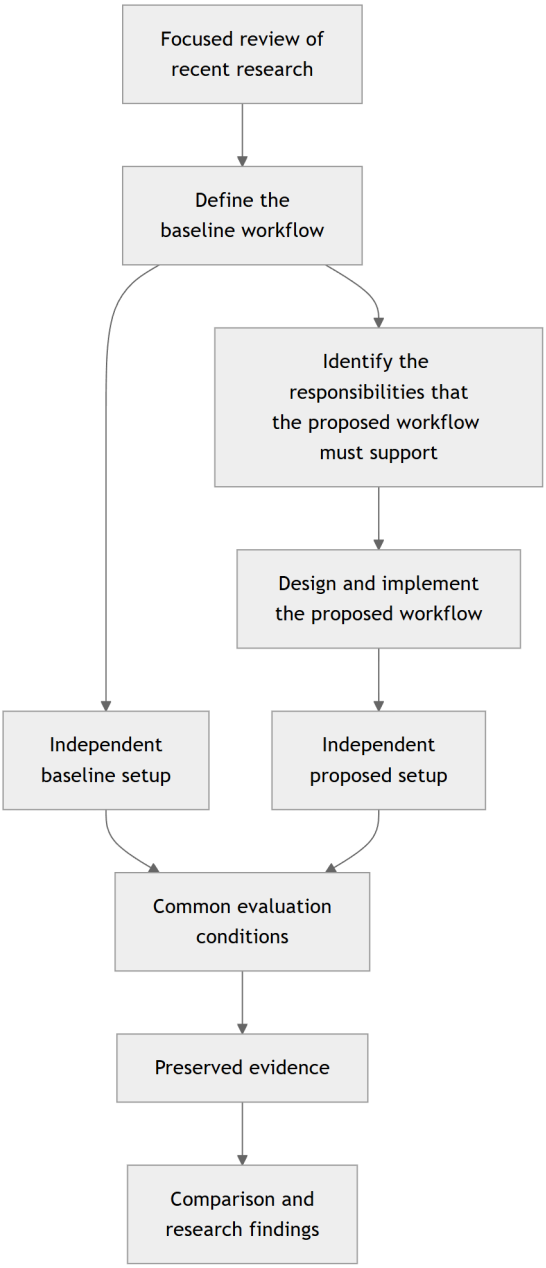


Figure 3.1: Research design from the focused literature review to the controlled comparison.

The proposed workflow was refined before the controlled comparison. The study reports one bounded design-and-evaluation cycle in one controlled case. Earlier informal refinements are not presented as separately evaluated cycles. Further iteration is left to subsequent projects; the study does not claim that one cycle establishes a mature or fully validated artefact [50].

The two evaluation branches remain separate. The baseline workflow is not deliberately weakened, and the proposed workflow is not allowed to use different

acceptance conditions. The intended difference is the way information is organised and made available during the work.

3.2 Unit of analysis and study boundary

The unit of analysis is the *scenario-to-simulator information workflow*. A unit of analysis is the main subject examined by a study.

This means that the research studies how information is interpreted, connected to simulator settings, checked, changed and preserved as evidence. The study does not treat the physical system represented by the simulator as its main subject.

CHESSCON output is used to assess the eight selected execution acceptance conditions. It is not used to compare or improve real-world operational performance.

The controlled comparison begins with an already configured model. The study therefore evaluates the information and verification work around that model. It does not measure the complete construction of a CHESSCON project from an empty file.

The study boundary prevents simulator output from being confused with workflow quality. A successful simulation can confirm that specified acceptance conditions were satisfied. It cannot, by itself, prove that every real-world detail is valid or that one information workflow is universally better.

3.3 Controlled comparison design

The baseline and proposed workflows were evaluated in separate setups created from the same verified reference model. The original project files matched the reference fingerprint. The retained baseline manifest was captured after an early controller opening and contained two rotated controller-metadata files; its other 177 files matched the reference. Therefore, configuration equivalence was checked explicitly, and byte-for-byte identity of the entire directories at accepted launch is not claimed.

Both setups used the same CHESSCON application, the same reference scenario and the same acceptance conditions. The simulator remained human-operated in both workflows. The proposed workflow did not automatically change or execute the CHESSCON project.

The treatment difference was the information available around the simulator. A *treatment* is the method or support supplied to one branch of a comparison. The study held the intended model conditions constant so that differences in the surrounding information work could be examined separately from differences in the simulated system. The existing model supplied a stable starting point, but also limited the study: decisions made during its original construction were not observed as part of either trial.

The baseline workflow used the written scenario information, simulator handbook, retained baseline description and project. Preparation involved identifying the required conditions in those materials and locating the corresponding configuration in CHESSCON. The proposed package was not consulted during baseline preparation. This preserved the direct source-to-simulator character of the baseline without withholding ordinary simulator guidance or the common acceptance rules.

The proposed setup used the controlled information records, mapping file, primary trace table, framework-audit check register and configuration guide developed in this thesis. These records supplied a documented path from a selected scenario condition to the simulator location and expected evidence. They did not perform the inspection: CHESSCON was opened and operated by a person in both setups. Consequently, the treatment compares available information support, not manual simulation against automated simulation.

Table 3.1 identifies what was shared and what differed. Independent project copies kept configuration and result files separate, while common acceptance conditions supplied the same basis for judging both outcomes. The original project fingerprint and the recorded metadata rotation are reported together because common origin is not identical to complete directory identity at every later point.

Table 3.1: Controlled factors in both setups.

Controlled factor	Baseline workflow	Proposed workflow	Reason for control
Starting configuration	Copy of the reference model; controller metadata rotation recorded	Separate copy of the same reference model	Common model origin with explicit configuration checks
Scenario	Same written scenario	Same written scenario	Ensures both workflows represent the same intended situation
Simulator	CHESSCON	CHESSCON	Removes software-product variation
Simulator operation	Human-operated	Human-operated	Prevents automation from becoming an uncontrolled difference
Project state	Independent setup	Independent setup	Keeps project and execution records separate
Acceptance conditions	Same conditions	Same conditions	Allows equivalent outcomes to be judged consistently
Workflow information	Normal source material and guidance	Organised workflow-support package	Defines the intended comparison
Evidence preservation	Same evaluator rules	Same evaluator rules	Allows both outcomes to be inspected

Project-identity checks, evidence preservation and post-execution acceptance rules were common evaluation controls applied to both setups. The workflow-support package also documented the expected identity and evidence responsibilities in advance. That recorded guidance is a property of the package; applying the common evaluation checks is not counted as an advantage of either workflow. The observed setup events therefore cannot establish that the proposed guidance prevented errors.

The observation record distinguished preparation, simulator interaction, execution, result verification and evidence handling. These categories describe different work: locating information is not the same activity as waiting for a simulation, and inspecting an output is not the same as preserving its original records. Preparation durations and setup events were retained, but complete matching start and finish points were

not available for every category. The analysis therefore reports the observed intervals and identifies unavailable measurements rather than filling missing categories with estimates.

An execution was assessed through the expected equipment identities, active-record counts and completed-job conditions defined in Chapter 5. A rejected setup event was not treated as an accepted trial. Similarly, a screenshot was not treated as numerical proof simply because it displayed a result screen. Original result-table and index pairs supplied the numerical basis; derived summaries made that basis readable. This evidence hierarchy avoids mixing configuration illustrations, unsuccessful setup events and accepted execution results.

The baseline trial preceded the proposed trial. Separate copies protect file state but do not remove knowledge gained during the earlier activity. This ordering and the researcher's familiarity with the package are retained as interpretation limits. They matter particularly for effort observations, where a shorter interval can have several explanations other than the information support itself.

Later changes are examined separately from the reference comparison. An executed change requires preserved simulator evidence. A change examined only through affected information and configuration locations is reported as impact analysis. The two evidence levels are not treated as equivalent.

3.4 Evidence and analysis

The study uses several forms of evidence because no single file can establish the source meaning, project identity, execution state and numerical result.

The evidence is arranged according to authority: the particular claim that a record can support. Table 3.2 separates those roles so that a source statement, a screenshot and an execution record are not treated as interchangeable proof.

Table 3.2: Evidence groups and their authority in the study.

Evidence group	Purpose in the study	Authority
Research literature	Supports the definition of the baseline and the design decisions	Basis for RQ1 and the design responsibilities
Scenario information	States what the configured model is intended to represent	Source of scenario meaning
Proposed-workflow records	Show the organisation supplied by the proposed workflow	Evidence of the designed artefact
Project-identity records	Connect evidence to the intended project and configuration	Evidence of configuration identity
Execution records	Connect output to a particular simulation execution	Evidence of execution history
Raw result table and companion index	Preserve stored result rows; table status identifies selected active rows and the companion index cross-checks their count	Primary numerical evidence when interpreted using the documented selection rule
Derived tables and charts	Present recalculated results in a readable form	Explanatory evidence derived from the raw records
Screenshots	Explain the CHESSCON interface and relevant locations	Interface evidence unless supported by controlled provenance

CHESSCON results are stored through a result table and a companion index. The retained reader selects rows by table status and cross-checks their count against the companion index. It does not traverse every index link. A physical row remaining in a result file is not automatically treated as part of the accepted output.

Derived tables and charts improve readability but do not replace the raw evidence. Every derived value must be recoverable from the retained source records.

The comparison first checks whether both executions satisfy the same eight acceptance conditions. The resulting claim is equivalence on the eight selected acceptance conditions. It does not establish equality of all simulator settings or preservation of every scenario property.

After equivalence is established, the workflows are compared across the separate software-quality dimensions shown in Table 3.3.

Table 3.3: Software-quality dimensions used in the comparison.

Evaluation dimension	Purpose
Functional correctness	Check the eight selected execution acceptance conditions.
Internal trace completeness	Determine whether included controlled records have explicit traces.
Framework-audit outcomes	Report agreement or managed uncertainty in the fifteen framework-audit checks.
Scenario-change handling	Determine which affected information categories are documented.
Data integrity	Determine whether active and inactive result records are interpreted correctly.
Evidence completeness	Assess evidence availability; disclose when a required-item denominator is missing.
Recorded effort observations	Report supported times and event counts, not a productivity estimate.

The definitions and acceptance rules are presented in Chapter 5. Data integrity and evidence completeness contribute to auditability, meaning the ability to examine the basis of a finding. Chapter 6 reports unavailable measures as well as supported observations, and maps the eight workflow requirements to their evidence.

The analysis is descriptive. Each dimension is reported separately using counts, ratios, classifications and supported observations. The dimensions are not combined into one overall score because such a score could hide important differences. No statistical comparison is used.

3.5 Alignment with the research questions

Table 3.4 shows how each Research Question is answered through the research method.

The RQ1 method does not begin by assuming that direct work is unsuccessful. It uses existing research to identify the responsibilities that a credible baseline should contain.

The RQ2 method converts those responsibilities into the design of the proposed workflow. The design is evaluated as an organised information solution around a human-operated simulator, not as an automatic importer.

The RQ3 method compares the workflows only after common acceptance conditions have been applied. Strong evidence in one evaluation dimension is not used to hide weaker or incomplete evidence in another.

Table 3.4: Alignment between the research questions and the research method.

RQ	Method	Main evidence	Intended output
RQ1	Focused literature synthesis followed by responsibility and workflow derivation	Literature findings and the synthesis presented in Chapter 2	A literature-supported baseline workflow
RQ2	Design of the proposed workflow using responsibilities identified from the research problem and literature	Workflow components, information relationships and responsibility boundaries presented in Chapter 4	A traceable, human-operated workflow design
RQ3	Descriptive comparison of two independent setups under common scenario and acceptance conditions	Measures defined in Chapter 5 and evidence reported in Chapter 6	A dimension-by-dimension comparison without a combined score

Detailed limitations and threats to validity are discussed in Chapter 7. The ethical and access conditions governing the study are stated in Section 1.5.

The method therefore connects the literature-derived baseline, the proposed workflow and the controlled comparison without presenting the results in advance. Chapter 4 defines the study scenario and explains the proposed workflow.

Chapter 4

Design and Implementation of the Proposed Workflow

4.1 Purpose of the chapter

Chapter 2 defined the baseline workflow. In its general form, a modeller reads scenario documents, interprets requirements, identifies simulator settings, configures and verifies the model, runs the simulation and records results. The accepted comparison exercised review and verification around an existing model, not all stages of new-model construction.

The literature review identified several concerns with this direct approach. Important values may be distributed across multiple documents. The relationship between a scenario requirement and a simulator setting may remain in the modeller's memory. A later change may require the same information to be found and interpreted again. The evidence used to confirm a setting or result may also be unclear.

This chapter presents the proposed workflow developed to address those concerns. The workflow adds a structured information layer between the written scenario and CHESSCON. It stores the scenario information, records how that information relates to CHESSCON, defines checks, prepares configuration instructions and preserves evidence.

The proposed workflow does not operate CHESSCON automatically. A person still selects the project, inspects the configuration, starts the simulation and reviews the results.

Table 4.1 connects the main concerns found in the baseline workflow with the corresponding design decisions.

Table 4.1: Connection between baseline concerns and proposed-workflow decisions.

Concern identified in the baseline workflow	Design decision in the proposed workflow
Scenario information may be distributed across several sources	Store the selected information in one controlled information model
The relationship between a requirement and CHESSCON may remain implicit	Record explicit scenario-to-CHESSCON mappings
Source information may be difficult to follow after configuration	Maintain a trace from the scenario source to the setting, check and evidence
Required checks may depend on the modeller's memory	Define the checks before configuration and execution
A later change may require repeated searching and interpretation	Record which information, mappings, CHESSCON locations and evidence are affected

Table 4.1 (continued)

Concern identified in the baseline workflow	Design decision in the proposed workflow
The wrong project may be opened	Require the complete working-copy path and project file to be confirmed
Stored result rows may not all be active	Interpret each result table together with its companion index
Screenshots and numerical evidence may be confused	Record the purpose and authority of every evidence item

The remainder of the chapter defines the scenario, explains the proposed workflow and describes how its information is used during CHESSCON configuration and verification. Evaluation outcomes are presented later and are not reported in this chapter.

4.2 Electric Forklift Movement Scenario

4.2.1 Purpose and scope

This thesis defines the *Electric Forklift Movement Scenario* as the bounded case for comparing the baseline and proposed workflows. Its defined conditions and the extent of their verification are reported separately.

The scenario represents a material-movement system in which battery-electric forklifts transport items between a ship-side transfer point and two storage areas. A ship-side transfer point is the location where items are received from or delivered towards a ship. The model does not attempt to represent an entire real terminal. It contains only the locations, equipment and activities required for the study.

The layout, movement flows, workload, equipment arrangement and charging conditions were designed as part of this thesis.

4.2.2 Scenario layout and movement

The scenario contains an outgoing storage area, an incoming storage area and a ship-side transfer point. The locations are connected through a road network used by the forklifts.

The outgoing storage area is identified inside the model as *Storage Area S1*. It is the starting location for outgoing items. The incoming storage area is identified as *Storage Area S3*. It is the destination for incoming items. These names are internal layout identifiers.

The scenario contains two movement directions. Incoming items move from the ship-side transfer point to Storage Area S3. Outgoing items move from Storage Area S1 to the ship-side transfer point.

Figure 4.1 presents the scenario structure.

The figure contains two connected movement flows. Both flows use the same fleet of four forklifts. The charging stations support the fleet but do not create or receive movement jobs. Figure 4.2 shows how this logical structure is represented in the CHESSCON layout.

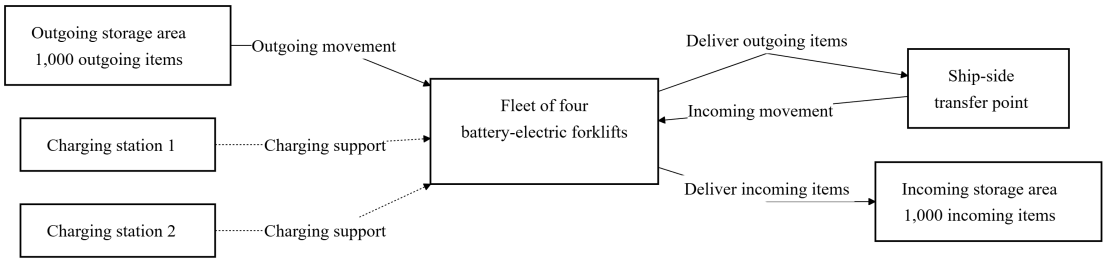


Figure 4.1: Locations, movement directions and equipment in the Electric Forklift Movement Scenario.

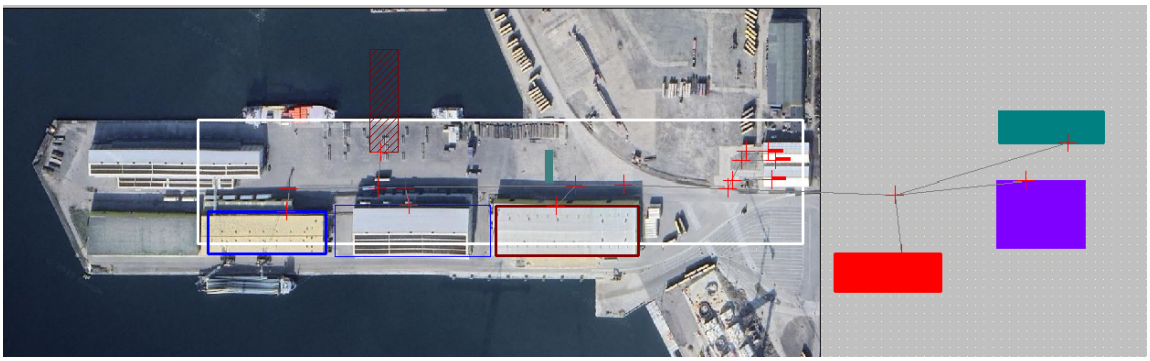


Figure 4.2: Two-dimensional CHESSCON layout of the scenario. The aerial background positions the working area, the outlined regions show modelled areas, and the connected lines show the route geometry used by the simulation. Author screenshot of CHESSCON, reproduced with permission from akquinet.

The layout image explains the spatial form of the model, while Figure 4.1 explains the meaning of the locations and movement directions. The screenshot contains no readable distance scale, so it is not used as evidence for the 50-metre scenario assumption.

4.2.3 Defined scenario conditions

One completed movement from its starting location to its destination is treated as one job. The scenario contains 1,000 incoming jobs and 1,000 outgoing jobs, giving a total workload of 2,000 jobs.

The incoming and outgoing quantities remain separate because they use different starting locations, destinations and item types. The combined total of 2,000 does not mean that the two movement directions are identical.

Table 4.2 summarises the defined scenario conditions.

Table 4.2: Defined conditions of the Electric Forklift Movement Scenario.

Scenario element	Defined condition	Meaning
Incoming workload	1,000 pallets	Movement from the ship-side transfer point to Storage Area S3
Outgoing workload	1,000 large bags	Movement from Storage Area S1 to the ship-side transfer point
Total workload	2,000 jobs	Complete movement requirement

Table 4.2 (continued)

Scenario element	Defined condition	Meaning
Unit weight	1 tonne	Weight assigned to each handled item
Forklift fleet	Four forklifts	Equipment available for both movement directions
Forklift model	Jungheinrich EFG 550	Equipment model represented in the scenario
Power source	Battery-electric	Forklifts use stored electrical energy
Battery capacity	74 kilowatt-hours	Maximum stored energy for each forklift
Lower charging threshold	20 per cent	Battery level at which charging may be required
Upper charging threshold	80 per cent	Intended battery level at which charging ends
Charging stations	Two	Charging capacity available to the fleet
Operating period	72 hours	Time represented by the simulation
Number of runs	One	Defined execution setting for the controlled comparison
Initial distance assumption	50 metres	Starting distance condition used in the scenario

The four forklifts form one shared fleet. The scenario does not assign a fixed number of jobs to each forklift. CHESSCON may distribute the work differently among them while completing the same total workload.

The two charging stations allow the forklifts to restore their stored energy when required. A forklift that is charging is temporarily unavailable for movement work. Charging is included because it affects equipment availability, but the thesis does not attempt to optimise the batteries or charging stations.

The 50-metre distance is a scenario assumption. It does not represent the total distance travelled by all four forklifts. Each forklift can complete many movements during the 72-hour operating period.

4.2.4 Scenario changes used to examine the workflows

Three changes extend the reference scenario so that change handling can be examined as well as the unchanged configuration. The workload change increases the number of incoming and outgoing jobs while leaving the equipment arrangement unchanged. It tests whether a revised quantity can be followed from the scenario into workload settings and new output expectations. The fleet change reduces the number of available forklifts while retaining the original work requirement. It tests a different relationship: an equipment type can remain unchanged even though the number of deployed units changes.

The distance change increases the stated movement distance. It tests whether the workflow distinguishes a scenario assumption from the route geometry represented in CHESSCON and the accumulated travel reported after execution. These are related concepts, but they are not the same measurement. This case is limited to identifying affected information and configuration responsibilities; no new simulated performance result is claimed for it.

Table 4.3 defines the three changes and their purpose. They provide change cases for examining the proposed records and the responsibilities of the baseline workflow. The preserved evidence does not contain a matched baseline and proposed execution

for every changed scenario. The workload and fleet cases have execution evidence, while the distance case remains an impact analysis. Their outcomes are reported in Chapter 6.

Table 4.3: Defined changes to the reference scenario.

Change	Revised condition	Relationship examined
Workload	1,200 incoming and 1,200 outgoing jobs instead of 1,000 in each direction.	Scenario quantity, workload mapping and expected completed-job totals.
Fleet	Three electric forklifts instead of four; 2,000 jobs remain required.	Equipment type, deployment count and active result records.
Distance	75 metres instead of the stated 50 metres.	Scenario distance, route representation and affected verification responsibilities.

4.3 Overview of the proposed workflow

The proposed workflow is designed to support scenario-to-CHESSCON configuration through connected information stages. Its evaluated use concerned reviewing an already configured model and checking selected results, rather than demonstrating complete conversion.

The controlled information model organises selected scenario statements, simulator context and evaluation evidence through stable identifiers and explicit relationships. It is a human-readable information structure, not an executable simulation model. The information is stored independently of the CHESSCON interface. Mappings then explain how the stored information relates to CHESSCON. Checks define what must agree. A configuration guide tells the modeller what to inspect. After execution, the results and evidence are connected back to the original scenario information.

Figure 4.3 shows the complete workflow and its responsibility boundary.

The diagram follows the information flow from source records to a configuration guide, human simulator use and preserved results. The package is stored as local files, not deployed as a service. The person transfers information across the CHESSCON boundary. The common evaluation tool reads exported files afterwards; it is not an interface that the package calls automatically.

The return arrow denotes a check against the workflow-support package, not an automatic update to CHESSCON.

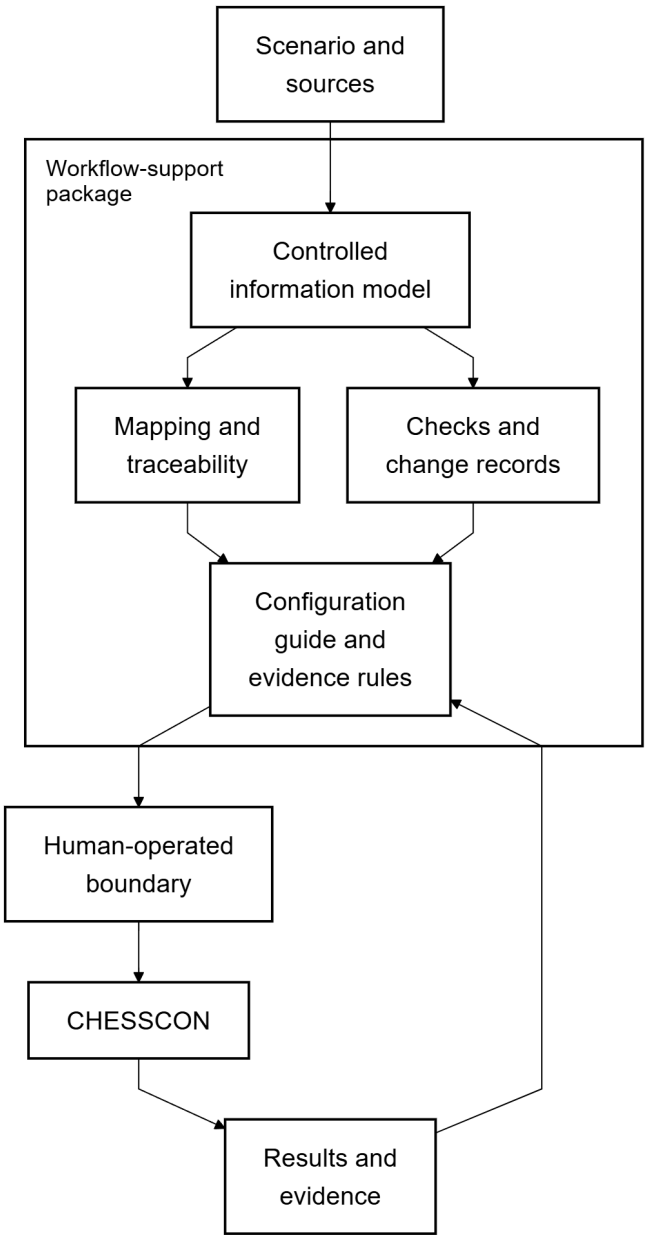


Figure 4.3: Information flow and responsibility boundaries in the proposed workflow.

4.3.1 Workflow-support package

Table 4.4 summarises the information stored in the workflow-support package.

Table 4.4: Information contained in the workflow-support package.

Package element	Format	Quantity	Main purpose
Controlled information records	YAML	30 records	Store selected controlled information, meanings and sources, including values and units where applicable
Scenario-to-CHESSCON mappings	YAML	Nine mappings	Explain how stored information relates to CHESSCON
Primary trace table	CSV	30 rows	Provide one readable source-to-target path for each record
Framework-audit checks	Markdown	15 checks	Define what information must agree
Configuration guide	Markdown	One guide	Give the modeller ordered CHESSCON instructions
Change-impact records	Markdown	Three cases	Identify what may be affected by a scenario change
Evidence catalogue	CSV	37 catalogue entries	Identify preserved evidence and its intended use

YAML is a human-readable structured text format. Its name, "YAML Ain't Markup Language", contains the acronym itself; this is called a recursive acronym. CSV means comma-separated values and is a plain-text table format. Markdown is a text format used for readable instructions and reports. These formats serve different purposes here: structured records hold values, tables expose links, and prose explains the human actions and interpretation rules.

These formats can be opened without specialised software-development tools. They do not replace the CHESSCON project files.

Retrospective explanation of the implemented design. The retained package separates information that needs its own meaning, source or review decision into controlled records. A required quantity, its equipment type and the identity of deployed units therefore have separate responsibilities even when they concern the same equipment. A mapping groups records around a simulator responsibility and its human action: the published deployment mapping, for example, combines fleet information needed to inspect deployment. A separate check states a distinct condition that can receive its own outcome and evidence-based explanation. These boundaries explain the implemented design; they are not a recovered formal selection protocol. The 30 records, nine mappings and fifteen checks describe the selected package, not an exhaustive inventory of all scenario requirements or possible checks.

4.4 Controlled information and mappings

4.4.1 Controlled information records

A controlled information record stores one clearly defined piece of information. Each record contains a stable identifier, readable label, source and intended meaning, together with controlled content or a value and unit where applicable.

The implemented records use identifiers beginning with SF, originally meaning "scenario fact". Its 30 records cover selected scenario conditions, simulator context and evaluation evidence; they are not 30 independent customer requirements. The original identifiers remain unchanged for auditability. Scenario records express selected conditions; context records identify the simulator and project; evidence records identify results and their interpretation. Table 4.5 defines identifier families, not a record-by-record classification of these categories. That complete classification and the unpublished check/evidence links remain in the private package; the PDF examples do not constitute a complete public trace.
Listing 4.1 shows a shortened YAML example for the required number of forklifts.

Listing 4.1: Shortened controlled information record for the required forklift fleet.

```
- id: SF21
  label: number_of_forklifts
  value: 4
  unit: forklifts
  source: Electric Forklift Movement Scenario
  meaning: four battery-electric forklifts are available
```

This record stores more than the number four. It states what the value represents, which unit applies and where the requirement was defined.

The record does not prove that four forklifts have already been configured in CHESSCON. It defines what the later configuration must represent.

4.4.2 Mapping records

A mapping record explains how one or more controlled records relate to CHESSCON. Mapping identifiers begin with MAP, meaning "mapping".

One scenario statement does not always correspond to one CHESSCON field. The requirement for four electric forklifts involves the fleet quantity, individual forklift records and the selected equipment type. These items must therefore be interpreted together.

Listing 4.2 shows the related shortened mapping.

Listing 4.2: Shortened mapping record for the forklift deployment.

```
- id: MAP006
  controlled_records: [SF21, SF22, SF23]
  chesscon_area: equipment deployment
  expected_state: four active forklifts using type FL-1
  human_action: confirm the deployment rows
  result_evidence: indexed forklift result table
```

FL-1 is the internal CHESSCON label for the forklift type used in the model. The mapping states where the related information is represented and what the modeller must confirm.

The mapping does not become the source of the number four. That value remains controlled by SF21. The mapping owns only the explanation of how the requirement relates to CHESSCON.

Table 4.5: Identifier families used in the published workflow description.

Identifier	Meaning in this thesis
`SF`	Stable controlled-record identifier; originally "scenario fact", also used for simulator context and evaluation evidence.
`MAP`	Mapping that relates controlled records to a simulator responsibility.
`CHK`	Framework consistency-check identifier; distinct from the eight execution acceptance checks.
`EV`	Evidence-catalogue entry identifier.
`WR`	Workflow Requirement identifier.
`C1-C3`	Workload, fleet and distance change cases, respectively.

4.4.3 Separation of responsibilities

Table 4.6 shows how the different information types remain separate.
Table 4.6: Separation of information responsibilities.

Information	Main responsibility
Scenario definition	States what the model must represent
Controlled record	Stores selected content and its meaning, with a value and unit where applicable
Mapping	Explains the relationship to CHESSCON
Configuration instruction	Tells the modeller what to inspect
Consistency check	Defines what must agree
CHESSCON project	Stores the product-specific configuration and results
Companion index	Supplies the live-count value used to cross-check status-selected result rows
Evidence record	Identifies what evidence exists and what it can support

This separation is intended to reduce competing definitions. Other records can refer to the controlled value through its identifier, but the package does not automatically enforce that rule or prevent copied values from diverging.

One mapping may use several controlled records. One controlled record may also participate in more than one mapping. The complete mapping file therefore contains more relationships than the 30-row primary trace table. The trace table provides one main review path for each controlled record, while the mapping file retains the additional relationships.

4.5 Applying the workflow in CHESSCON

4.5.1 Preparing the configuration guide

The controlled records and mappings are converted into an ordered configuration guide. In this context, *converted* means rewritten as human-readable instructions. It does not mean automatic transfer into CHESSCON.

Each instruction contains the information shown in Table 4.7.

Table 4.7: Information contained in a configuration instruction.

Instruction information	Purpose
Related record and mapping	Identifies the scenario information being implemented
CHESSCON location	Tells the modeller where to work
Expected condition	States what should be visible
Human action	Explains what must be inspected or configured
Evidence requirement	States what must be preserved
Stop condition	States when the activity must not continue

For example, the fleet instruction directs the modeller to the equipment-deployment table, states that four entries using type `FL-1` are expected and requires four active forklift result rows after execution.

4.5.2 Human-operated CHESSCON boundary

The workflow-support package cannot directly read from or write to CHESSCON. A person acts as the boundary between the package and the simulator. The evaluation reader is separate software, applied to both workflows after execution.

Table 4.8 separates the responsibilities.

Table 4.8: Responsibilities at the human-operated CHESSCON boundary.

Activity	Proposed-workflow responsibility	Human or CHESSCON responsibility
Project selection	State the expected path and project identity	The modeller opens and confirms the project
Configuration review	State the expected values and locations	The modeller reads the visible settings
Configuration entry	Provide the ordered instructions	The modeller enters authorised values
Start decision	State the required checks and stop conditions	The modeller starts or cancels the run
Execution	State the required run conditions	CHESSCON executes the model
Result selection	State the required result family and index rule	The modeller opens the results; CHESSCON stores the data
Acceptance	Define the comparison condition	The modeller records the decision from the evidence
Preservation	Define the required evidence	The evidence files are retained for review

The boundary prevents the workflow from being presented as an automatic importer. The modeller remains responsible for applying the instructions correctly.

4.5.3 Project identity

Before configuration or execution, the complete working-copy path and main project file must be confirmed. The main CHESSCON project file uses the `.TER` file extension. A file extension is the final part of a filename that indicates its type.

A familiar project name is not sufficient because different folders may contain similar names. If the visible path does not point to the intended working copy, the activity must stop.

The workflow may also record a file hash. A hash is a calculated digital fingerprint used to identify a file. A matching hash supports project identity but does not prove that every visible setting is correct.

4.5.4 Before-execution checks

The configuration guide identifies the conditions that must be inspected before execution. Table 4.9 summarises the checks for the Electric Forklift Movement Scenario.

Table 4.9: Conditions inspected before execution.

Configuration area	Expected condition
Project	Intended working-copy path and project file
Forklift type	Battery-electric type `FL-1`
Fleet deployment	Four forklift entries
Incoming workload	1,000 jobs
Outgoing workload	1,000 jobs
Operating period	72 hours
Number of runs	One
Ship-side transfer point	Intended active transfer record
Charging facilities	Two charging-station records

These checks inspect selected visible conditions. They do not establish every property in the scenario definition or prove that execution will complete successfully. Table 6.6 records the available coverage.

4.6 Verification, evidence and change handling

4.6.1 Indexed-result verification

After execution, the evaluation procedure reads exported table/index pairs with a retained Python tool. It selects table records carrying the recognised active-status value and compares their count with the companion index's live-count field. It does not traverse the index's complete row mapping. Section 5.8.1 documents this common evaluation support, which is separate from the workflow-support package and was applied to both accepted executions.

The companion index contains result-selection information. The retained reader uses only its live-count field, together with table statuses. A physical table may retain an inactive older row, so counting every physical row could produce an incorrect result.

Figure 4.4 shows the verification path.

The CSV summary improves readability, but it does not replace the original table and index. Both original files must be retained.

4.6.2 Evidence preservation

Different evidence types support different claims. Table 4.10 defines their roles.

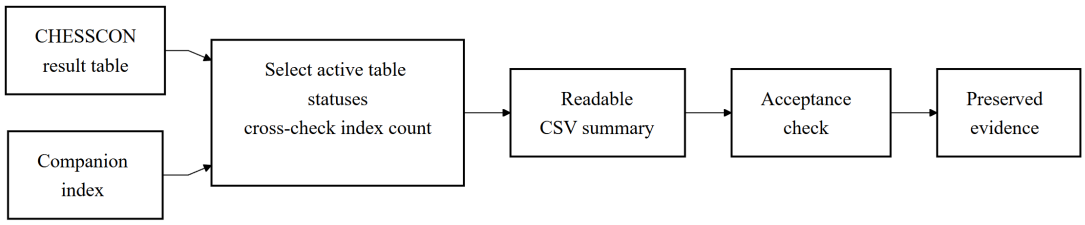


Figure 4.4: Status-based result selection and index-count checking applied to both accepted executions.

Table 4.10: Permitted use of the evidence retained by the proposed workflow.

Evidence type	Permitted use
Project path and filename	Identify the working copy
File hash	Support file identity
Configuration screenshot	Explain where a CHESSCON value is displayed
Raw result table	Preserve product-specific numerical records
Companion index	Cross-check the active count selected by table status
Parsed CSV summary	Support readable recalculation
Timestamp	Place an activity in its recorded sequence
Decision record	Explain why a result was accepted, rejected or treated as a managed warning

A screenshot can explain the CHESSCON interface. It is not automatically the strongest evidence for a numerical result. The indexed result table is used as the main source for numerical claims.

4.6.3 Framework-audit checks

A consistency check compares related information and records the outcome. Check identifiers begin with `CHK`, meaning "check".

The proposed workflow uses the three statuses defined in Table 4.11.

Table 4.11: Meaning of the framework-audit check statuses.

Status	Meaning
Pass	The stated evidence satisfies the condition
Managed warning	A known uncertainty exists, but its effect has been bounded and explained
Fail	The condition is not acceptable for the intended claim

A managed warning is not the same as a pass. It records a limitation that must remain visible when the evidence is interpreted.

The results of the fifteen framework-audit checks are reported in Chapter 6. These checks are distinct from the eight execution acceptance checks applied to each accepted result.

4.6.4 Change-impact records

A change-impact record identifies what may need to be reviewed after an approved scenario value changes. It does not perform the change automatically.

Table 4.12 shows the information contained in a change-impact record.
Table 4.12: Information contained in a change-impact record.

Change information	Purpose
Original value	Records the starting condition
Revised value	Records the approved update
Affected controlled records	Identifies which stored scenario information changes
Affected mappings	Identifies which CHESSCON relationships require review
Affected CHESSCON areas	Shows where implementation work may be required
Affected checks	Identifies which conditions must be repeated
Required evidence	States what new evidence must be preserved
Execution requirement	States whether another simulation must be performed

The later evaluation applies this structure to a workload change, a fleet change and a distance change. Chapter 6 reports which changes were executed and which were assessed only for impact.

4.7 Implementation scope

The proposed JSON (JavaScript Object Notation) importer was not implemented. Instead, a person uses the readable YAML, CSV and Markdown package. The common evaluator's `acceptance.json` summaries are separate evidence records, not a transformation interface. Table 4.13 distinguishes the implemented support from unimplemented automation.
Table 4.13: Boundary between the implemented workflow and unimplemented automation.

Implemented in the thesis	Not implemented
Controlled information	Automatic import into CHESSCON
Explicit scenario-to-CHESSCON mappings	Direct software connection to CHESSCON
Primary trace table	Automatic enforcement of the mappings
Framework-audit checks	Automatic correction of incorrect values
Ordered configuration guide	Automatic simulator execution
Change-impact records	Automatic propagation of scenario changes
Evidence catalogue	Automatic operation by the workflow-support package

These files support review but cannot prevent incorrect project selection, value entry or an ignored warning. Instructions, stop conditions and evidence checks manage those risks. Chapter 5 defines the evaluation of this human-operated workflow around the already configured model.

Chapter 5

Evaluation Design and Measures

5.1 Purpose of the evaluation

The evaluation was designed to compare the baseline workflow with the proposed workflow under the same conditions.

Both evaluations used independent copies of an already configured model for the Electric Forklift Movement Scenario and the same result requirements. The comparison covered preparation for review, configuration verification, execution and documentation. It did not measure construction of a model from an unstructured scenario. The supplied workflow material differed; the common evaluation procedure and evidence reader were not components unique to the proposed workflow.

The baseline workflow used the written scenario and supporting documents directly. The proposed workflow used controlled information records, mappings, trace links, predefined checks, configuration guidance and evidence rules.

The evaluation assessed RQ3 through four principal dimensions: modelling effort, traceability, consistency and handling scenario changes. Functional correctness, data integrity and evidence completeness supported interpretation of those dimensions. These are analytical dimensions chosen for the comparison, not additional research questions. The chapter defines their intended measures; the Results chapter distinguishes calculable observations from measures for which the necessary evidence was not retained.

No answers to the research questions are presented in this chapter. This chapter defines how the evidence was measured. Chapter 6 presents the measured values, findings and research-question answers.

5.2 Structure of the comparison

Figure 5.1 shows the complete comparison design.

The diagram contains baseline-workflow and proposed-workflow branches. Both start from the same configured model. They differ in the information supplied for reviewing, checking and documenting its representation.

Table 5.1 separates the changed factor, controlled factors and measured outcomes.

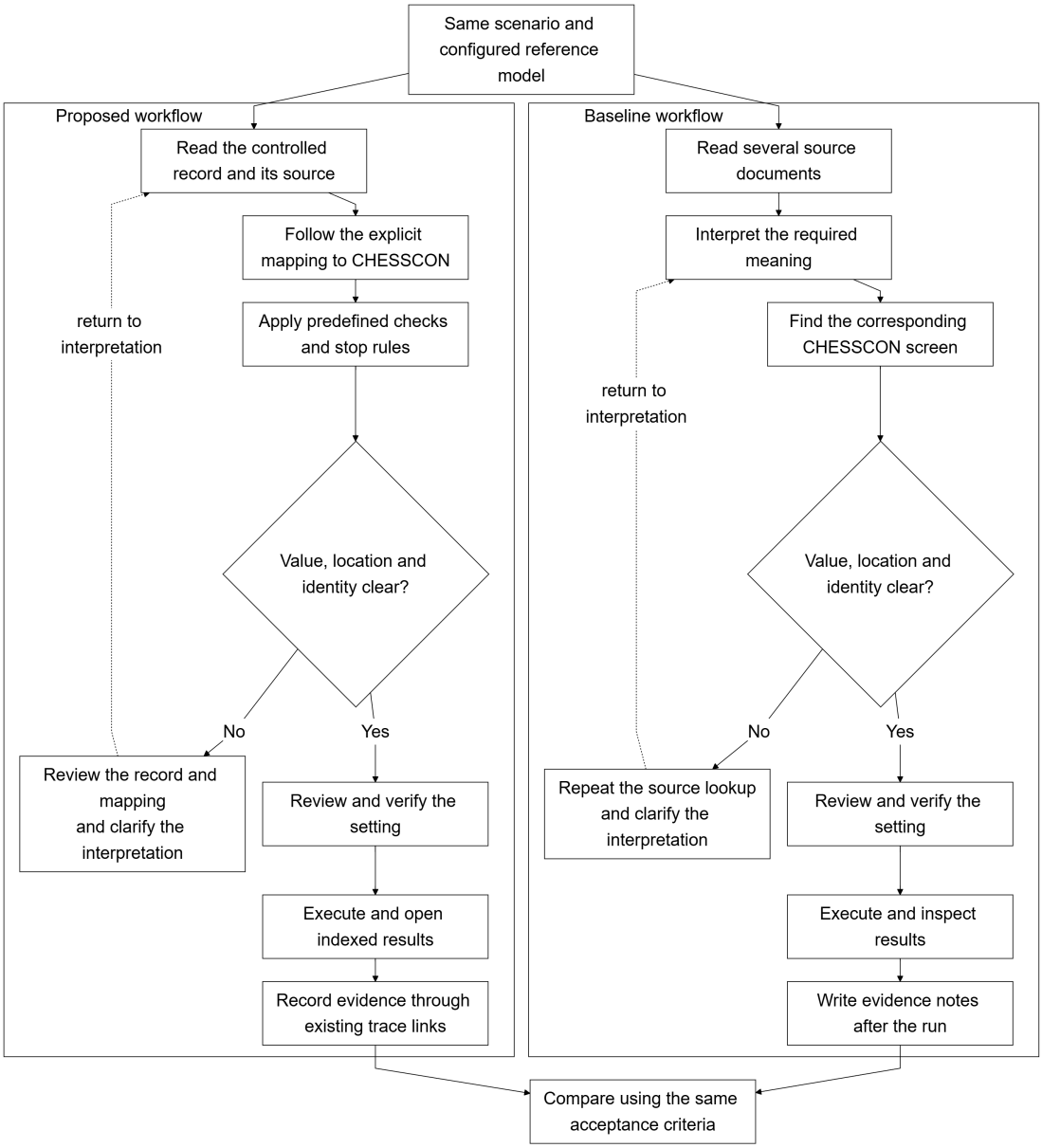


Figure 5.1: Comparison structure used in the evaluation. Both decision loops show possible review and correction, not observed frequencies or durations of rework. The difference is the information consulted, not an assumption that one workflow avoids repetition.

Table 5.1: Variables used in the workflow comparison.

Variable type	Evaluation item	Treatment
Changed factor	Workflow	Baseline workflow compared with proposed workflow
Controlled factor	Scenario	Same Electric Forklift Movement Scenario
Controlled factor	CHESSCON model	Independent copies of the same configured model
Controlled factor	Simulation duration	72 simulated hours
Controlled factor	Number of simulation runs	One accepted run for each workflow

Table 5.1 (continued)

Variable type	Evaluation item	Treatment
Controlled factor	Required workload	2,000 movement jobs
Controlled factor	Required equipment	Four battery-electric forklifts
Controlled factor	Acceptance conditions	Same conditions applied to both workflows
Measured outcome	Modelling effort	Recorded preparation and workflow time
Measured outcome	Traceability	Internal trace completeness for the controlled records
Measured outcome	Consistency	Pass, managed-warning and failure rates
Measured outcome	Change handling	Coverage of affected information
Measured outcome	Functional correctness	Agreement between expected and indexed CHESSCON results
Measured outcome	Evidence completeness	Availability of required supporting evidence
Measured outcome	Data integrity	Physical and status-selected counts; index-count agreement

Independent model copies were used so that files generated during one evaluation did not replace those from the other. Technical identifiers and file fingerprints remain in the retained execution records. Derived results are presented in the thesis; original licensed project files and restricted evidence are not publicly distributed.

5.3 Evaluation dimensions

Each evaluation dimension answered a different question about the workflow. The dimensions were measured separately to prevent one result from being used to support an unrelated claim. For example, a shorter preparation time would not automatically prove better traceability. Similarly, a complete trace table would not prove that CHESSCON produced a correct result.

Figure 5.2 shows how the dimensions are connected to the evidence.

Table 5.2 defines the meaning of every dimension.

Table 5.2: Definition of the evaluation dimensions.

Dimension	Meaning in this evaluation	Main measurement
Functional correctness	Whether the CHESSCON result satisfies the eight selected scenario acceptance conditions	Job-completion rate, active equipment count and required object identities
Modelling effort	Recorded work involved in preparing and verifying the scenario representation	Preparation time, interaction time, verification time and recorded corrections
Internal trace completeness	Recorded links within the selected information package	Percentage of its controlled records with an explicit trace
Consistency	Agreement between related values and evidence	Number and percentage of passes, managed warnings and failures
Scenario-change handling	Extent to which the five predefined impact categories are explicitly documented for a scenario update	Percentage of required impact categories documented

Table 5.2 (continued)

Dimension	Meaning in this evaluation	Main measurement
Data integrity	Correct separation of active results from stored but inactive records	Active, physical and inactive row counts
Evidence completeness	Availability of the evidence required to support an evaluation claim	Percentage of required evidence items available

5.4 Functional-correctness measures

Functional correctness was assessed only against the eight output conditions below. The retained evidence reader selected table records with status 256 and checked that their count agreed with the live-count field in the companion index. It did not traverse the complete index to select records. The term indexed results in this thesis refers to this status-based selection with an index-count cross-check, not an independent reconstruction of every index link.

Figure 5.3 shows the measurement path.

5.4.1 Total completed jobs

The total number of completed jobs was calculated by adding the job count of every active forklift record:

$$J_{\text{total}} = \sum_{r \in I_{\text{forklift}}} J_r \tag{5.1}$$

In Equation 5.1, I_{forklift} is the set selected by table status after the index-count cross-check, r identifies one selected record, J_r is its completed-job count and J_{total} is the sum across those records. Both J_r and J_{total} are measured in completed jobs. The summation includes each selected forklift once. The preserved records support this calculation.

The job-completion rate was calculated as:

$$C_{\text{jobs}} = \frac{J_{\text{total}}}{J_{\text{expected}}} \times 100 \tag{5.2}$$

Here, C_{jobs} is the completion percentage and J_{expected} is the required number of jobs for the evaluated scenario. The numerator and denominator are both job counts, and the denominator must be greater than zero. A value of 100% means that the selected forklift records completed the required total; it does not measure completion speed or equality of individual allocations.

5.4.2 Active fleet size

The active fleet size was calculated as:

$$N_{\text{active}} = |I_{\text{forklift}}| \tag{5.3}$$

In Equation 5.3, N_{active} is the active forklift count and the vertical bars mean the number of records in the set I_{forklift}. One selected row represents one active forklift.

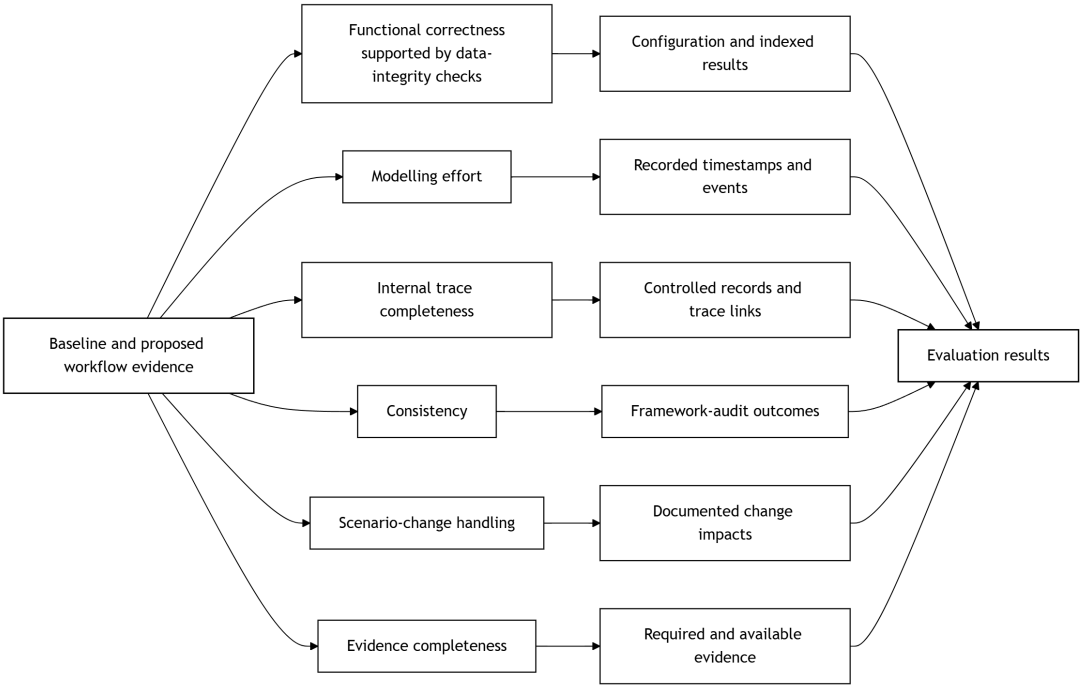


Figure 5.2: Measurement model used in the evaluation.

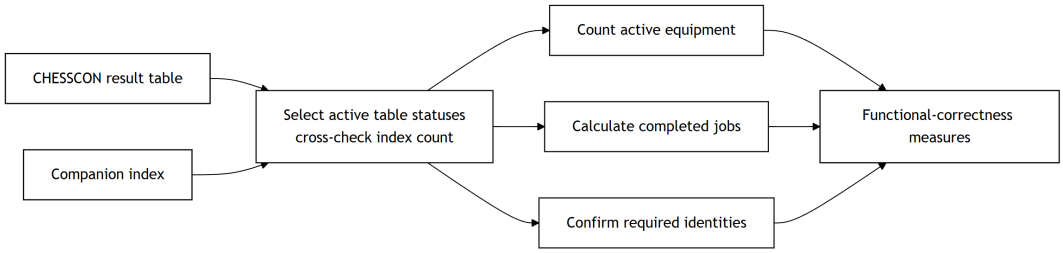


Figure 5.3: Selection and measurement of active CHESSCON results.

The fleet condition is satisfied when that count matches the fleet size defined by the evaluated scenario.

5.4.3 Eight execution acceptance checks

The eight acceptance checks assess counts, identities and workload fields that can be checked consistently in both retained output sets. Their expected values reflect the defined reference scenario and required object identities. Acceptance is limited to these selected properties rather than complete semantic equivalence; Table 6.6 identifies conditions not covered. The active records were assessed through the eight true-or-false checks in Table 5.3. The execution record separately noted whether a simulation error was observed; it was not counted as a ninth execution acceptance check.

Table 5.3: Eight execution acceptance checks applied to each unchanged-scenario evaluation.

No.	Check	Pass condition
1	Active forklifts	Four status-selected forklift rows; count agrees with the index
2	Forklift type	Every active forklift row uses type `FL-1`
3	Completed workload	Active forklift jobs sum to 2,000
4	Active ship-side transfer rows	One status-selected row; count agrees with the index
5	Ship-side transfer identity	The active row uses the expected internal identity `QC-1`
6	Transfer jobs and targets	Incoming, outgoing and both target values equal 1,000
7	Active charging rows	Two status-selected rows; count agrees with the index
8	Charging identities	The active names are `CHARGER-1` and `CHARGER-2`

The same eight checks were applied to both workflows.

5.5 Modelling-effort measures

Modelling effort was divided into the time categories in Table 5.4. The categories separate simulator calculation time from the human work of understanding and verifying the model. Their definitions also make missing observations visible: a category cannot be reported merely because an activity probably took place.

Table 5.4: Time categories used in the evaluation.

Time category	Start point	End point	Purpose
Preparation time	Beginning of workflow preparation	Scenario information ready for CHESSCON review	Intended preparation measure
CHESSCON interaction time	First relevant CHESSCON interaction	Simulation ready to start	Measures work in the simulator
Simulation time	Simulation start	Simulation completion	Records CHESSCON execution separately
Verification time	Result inspection begins	Acceptance decision is recorded	Measures result checks
Evidence-handling time	Evidence packaging begins	Final evidence is preserved	Records packaging work
Total tracked workflow time	Beginning of preparation	Final evidence is preserved	Represents the complete interval when all timestamps are available

The duration of one activity was calculated as:

$$T_x = t_{\text{end},x} - t_{\text{start},x} \tag{5.4}$$

Here, x identifies an activity category, $t_{\text{start},x}$ and $t_{\text{end},x}$ are its recorded start and finish points on the same time basis, and T_x is the elapsed duration in seconds. The calculation requires both points. Elapsed total workflow time similarly requires recorded start and finish points:

$$T_{\text{elapsed}} = t_{\text{final evidence}} - t_{\text{preparation start}} \tag{5.5}$$

In Equation 5.5, T_{elapsed} is the total elapsed duration in seconds, $t_{\text{preparation start}}$ is the beginning of preparation and $t_{\text{final evidence}}$ is the point at which final evidence preservation ends. Summing activity durations gives the same total only when all categories cover this interval without gaps or overlap. Any excluded waiting or diagnostic activity must be identified separately. Complete matching boundaries were unavailable, so no total workflow-time comparison is reported.

The retained preparation observations were labelled as preparation in the event record, but detailed activity logs were not frozen for both trials. Their correspondence to the intended boundaries in the table cannot therefore be established fully. Controller diagnostics and user waiting were explicitly excluded from recorded workflow timing. Simulator output-write windows are also distinct from complete simulator runtime.

The two retained preparation intervals are reported individually as pilot observations. No percentage reduction or absolute time-saving estimate is used to infer workflow performance. One observation per workflow and incomplete matching activity boundaries do not establish a reduction in total modelling effort.

Table 5.5 defines the intended additional counts. Some were not instrumented sufficiently; Table 6.11 distinguishes reported values from unavailable measures.

Table 5.5: Additional modelling-effort measures.

Measure	Calculation
Rejected starts	Number of starts cancelled before an accepted execution
Configuration corrections	Number of recorded configuration changes made before acceptance
Repeated lookups	Number of explicitly recorded repeated searches for the same information
CHESSCON areas consulted	Number of distinct simulator areas inspected
Evidence-processing actions	Number of recorded actions required to preserve and verify evidence

Only values supported by the recorded evidence are reported in Chapter 6. Missing values are not estimated.

5.6 Internal trace completeness

Internal trace completeness measures recorded relationships within the selected package. It does not measure how completely the package captured all information in the original sources.

The package contained 30 controlled information records, including scenario values, context and evidence-related information. These selected records formed the denominator:

$$C_{\text{trace}} = \frac{N_{\text{traced}}}{N_{\text{selected}}} \times 100 \tag{5.6}$$

In Equation 5.6, N_{traced} counts controlled records with the specified internal trace, N_{selected} is the count of controlled records selected for this measure (30), and C_{trace} is internal trace completeness in per cent. Each record is counted once, even if it participates in several mappings. No independent inventory of all source requirements was established. Source-capture coverage therefore has no defensible denominator and is not reported.

A record was counted as traced only when its trace contained all elements shown in Table 5.6.

Table 5.6: Elements required for an explicit trace.

Trace element	Required information
Controlled record	Stable identifier and controlled content or value
Source	Location from which the information originated
Mapping	Relationship between the controlled information and CHESSCON
CHESSCON responsibility	Simulator area or configuration responsibility
Verification target	Result or evidence used to confirm the representation

Evidence notes created during the evaluation were not counted as baseline workflow traceability. The measure considered only traceability that existed as part of the workflow being evaluated.

Mapping count and internal trace completeness were reported separately. A mapping can connect several controlled records, while the primary trace table provides one main review path for each required record.

5.7 Consistency measures

Consistency support was assessed using the retained proposed-workflow register of fifteen framework-audit checks. Table 5.7 defines its three statuses. Distinguishing a managed warning from a pass keeps unresolved interpretation issues visible. This framework audit covers the retained scenario and change evidence; it is separate from the eight execution acceptance checks applied to each accepted execution.

Table 5.7: Consistency-check classifications.

Status	Measurement meaning
Pass	The available evidence satisfies the defined condition
Managed warning	A known uncertainty is identified, bounded and kept visible
Fail	The required condition is not satisfied

The three outcome rates were calculated as:

$$R_{\text{pass}} = \frac{N_{\text{pass}}}{N_{\text{checks}}} \times 100, \tag{5.7}$$

$$R_{\text{warning}} = \frac{N_{\text{warning}}}{N_{\text{checks}}} \times 100, \tag{5.8}$$

$$R_{\text{fail}} = \frac{N_{\text{fail}}}{N_{\text{checks}}} \times 100. \tag{5.9}$$

The counts N_{pass}, N_{warning} and N_{fail} are the numbers of checks in each class. N_{checks} is the total number of classified checks, which is 15 in the retained register. The corresponding R variables are percentages. Each check receives one status, so the counts must satisfy:

$$N_{\text{pass}} + N_{\text{warning}} + N_{\text{fail}} = N_{\text{checks}} \tag{5.10}$$

The comparison examined whether predefined checking guidance was available within each workflow. The workflow-support package supplied this register; no equivalent consolidated baseline register was preserved. Its status percentages therefore describe the workflow-support package rather than a paired error-rate comparison. The eight common execution checks were applied to both accepted runs. Guidance available before execution may assist preparation, but its availability alone does not demonstrate error prevention.

5.8 Data-integrity measures

Data-integrity analysis examined whether the selected rows and their count were consistent with the retained execution evidence. The number of stored but inactive records was calculated as:

$$N_{\text{inactive}} = N_{\text{physical}} - N_{\text{active}} \tag{5.11}$$

Here, N_{physical} is the stored record count in the table under examination, N_{active} is its status-selected count checked against the index, and N_{inactive} is the excluded record count. All three quantities are numbers of records from the same table. The physical count must be at least the active count. For the forklift table, the active count is also the fleet count in Equation 5.3.

A difference between physical and active counts did not automatically make a result invalid. Table 5.8 sets out the checks needed before interpreting that difference: the files must belong together and the selected count must agree with the index.

Table 5.8: Data-integrity checks.

Check	Purpose
Result table exists	Confirms that the product-specific numerical records were preserved
Companion index exists	Supplies a live count for checking status-based selection
Table and index belong together	Prevents records from different executions being combined
Active-row count is recorded	Supports fleet and equipment verification

Table 5.8 (continued)

Check	Purpose
Inactive rows are excluded	Prevents older records from changing the result
Original files are preserved	Allows later audit of the parsed values
Readable summary is reproducible	Allows the selected rows and totals to be checked again

5.8.1 Evidence reader and verification boundary

The common evaluation procedure used the retained Python file `collect_evaluation_evidence.py`. It is an evidence-processing tool, not an executable part of the workflow-support package. Its SHA-256 identity is retained with the evaluation documentation. SHA-256 is a cryptographic fingerprint used to detect changes in a file; it does not prove that the file is correct. The recalculation used Python 3.11.9 on Windows; the original execution-time Python version was not preserved. The reader uses Python's standard library and consumes preserved result-table and index pairs. It produces readable record summaries and JavaScript Object Notation (JSON) acceptance records. JSON is a structured data format here, not the abandoned importer prototype.

The reader skips a 16,384-byte table header and reads fixed-length records: 376 bytes for forklifts, 224 for the ship-side transfer object and 128 for charging stations. It selects records whose unsigned status field at byte offset 8 equals 256. It compares their number with the unsigned two-byte live count at offset 8,201 in the companion index. Names use the Windows-1252 character encoding. A byte offset identifies a position within the stored data. These details describe the observed format, not a published CHESSCON interface contract. Table 5.9 records the checks used to examine the reader and its inputs.

Table 5.9: Checks on the retained evidence reader.

Check	Interpretation
Table alignment	The payload must contain a whole number of fixed-size records.
Malformed input	Truncated tables, short indexes and invalid text are rejected rather than silently repaired.
Active count	Status-selected record count must match the index live-count field.
Stale-record exclusion	In the reduced-fleet table, the retained inactive forklift row is excluded; three active rows remain.
Saved-summary agreement	Recalculated values are compared with the preserved summaries and acceptance records.

For a worked preserved example, the reduced-fleet table has four physical records. The status-based selection retains the three forklift records with 667, 728 and 605 jobs. Their count agrees with the index value of three and their total is 2,000. The retained row named `FL-002`, an individual forklift identifier, is excluded by its table status. Agreement with saved summaries checks repeatability of this decoding procedure. It is not independent validation against the CHESSCON application. In particular, matching counts cannot prove that the selected identities match every link in the full index. The QA record documents these checks and their limits without rerunning CHESSCON.

Here, `table` and `index` are Python path objects pointing to the preserved files; `376` is the record size in bytes and `flt` selects forklift fields. The collection modes that copy or replace evidence are not required for this inspection.

```
decoded = decode_records(table, index, 376, "flt")
active_rows = decoded["live_rows"]
```

The retained local quality-assurance (QA) record is named `verification_results.json`; the private evidence-location guide identifies its context. The exact historical CHESSCON build and Windows operating-system version were not retained in the inspected execution record. A current installation version would not establish those historical versions. Section 6.3.1 distinguishes the retrospectively reported computer details from the unavailable execution-time version record. The fixed-offset reader is therefore documented only for the preserved table format, not as a version-independent CHESSCON interface.

5.9 Evidence-completeness measures

Evidence completeness was defined as the following intended measure:

$$C_{\text{evidence}} = \frac{N_{\text{available}}}{N_{\text{required}}} \times 100 \tag{5.12}$$

In Equation 5.12, C_{evidence} is the intended evidence-completeness percentage, $N_{\text{available}}$ is the number of required evidence items preserved, and N_{required} is the number in a predefined required-item inventory. These are evidence-item counts, not the controlled-record counts used in Equation 5.6. No complete frozen evidence inventory was retained, so this percentage cannot be calculated defensibly. The 37-entry catalogue is an observed inventory size, not evidence of 100% completeness.

Table 5.10: Evidence categories used in the evaluation.

Evidence category	Purpose
Scenario evidence	Establishes the expected values and meanings
Workflow evidence	Shows the information available under the evaluated workflow
Configuration evidence	Supports the CHESSCON representation
Raw result table	Preserves the original numerical records
Companion index	Cross-checks the count of status-selected records
Parsed result summary	Supports readable recalculation
Timestamp evidence	Supports time measurements
Acceptance record	Records whether the defined conditions were satisfied
Change-impact record	Identifies what is affected by a scenario update
Evidence catalogue	States what evidence exists and what claim it supports

Table 5.10 separates evidence by purpose rather than assigning equal authority to every item. Screenshots were classified as configuration or interface evidence, not the primary source of controlled numerical claims when result tables were available.

5.10 Evaluation of scenario changes

Section 4.2.4 defines the workload, fleet and distance changes. Their evaluation follows two evidence paths, shown in Figure 5.4. The workload and fleet cases require both a documented change scope and a changed execution result. The distance case examines affected information only. Keeping these paths separate prevents an impact assessment from being presented as an executed performance experiment.

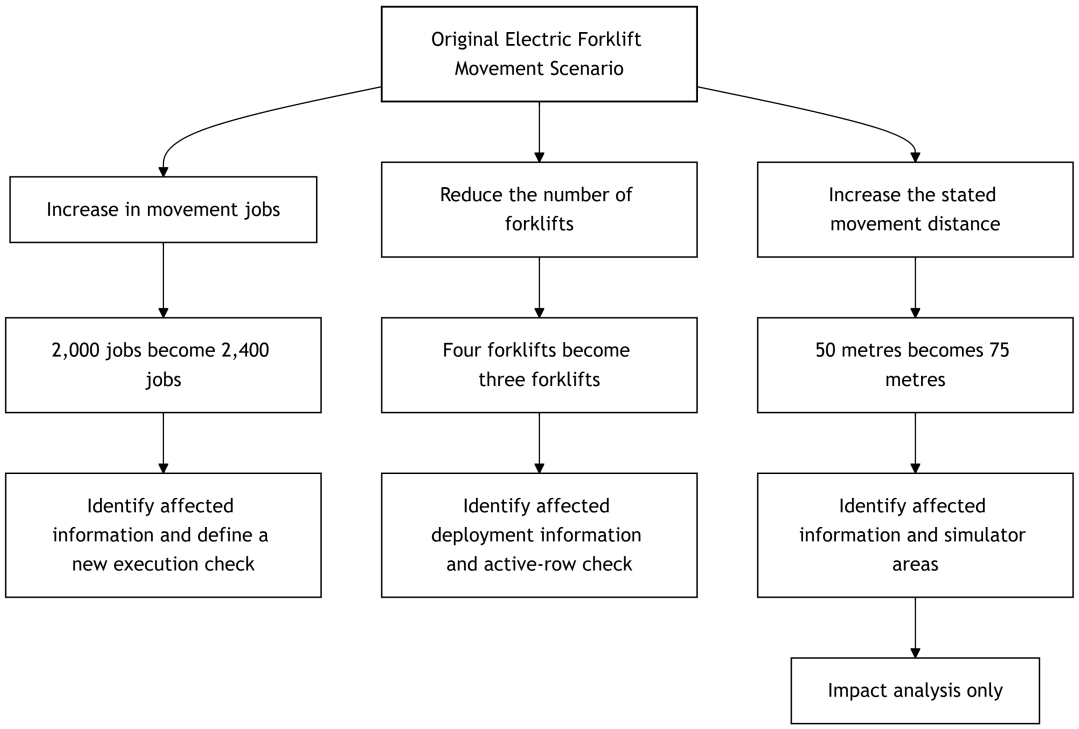


Figure 5.4: Scenario updates used to evaluate change handling.

For an executed change, the expected values in the acceptance checks are revised to match the changed requirement. The workload case changes the required job totals, while the fleet case changes the active forklift count and retains the original job total. The analysis then checks the corresponding preserved records. For the distance case, the evidence is the named information and simulator areas that would require review. No output total is inferred for a simulation that was not executed. None of these cases is treated as a matched pair of baseline and proposed change trials.

5.11 Change-impact measures

Change handling was evaluated by checking whether five required impact categories were documented: changed scenario information, affected mappings, affected CHESSCON areas, checks requiring repetition and required evidence.

For this revision, the preserved change records were assessed retrospectively using:

$$C_{\text{change}} = \frac{N_{\text{documented categories}}}{N_{\text{required categories}}} \times 100 \tag{5.13}$$

Here, C_{change} is the document-coverage percentage for one change case, $N_{\text{documented categories}}$ is the number of categories explicitly addressed in its record, and $N_{\text{required categories}}$ is five. Both counts refer to the categories in Table 5.11; each category is counted once. A generic reference to review was not counted as a named check. These are retrospective document-coverage scores, not measurements collected during execution or evidence of reduced change effort.

Table 5.11: Change-impact evaluation matrix.

Impact category	Question applied to each update
Scenario information	Which controlled values change?
Mappings	Which scenario-to-CHESSCON relationships are affected?
CHESSCON areas	Which simulator areas may require review?
Checks	Which conditions must be repeated or updated?
Evidence	What evidence is needed to verify the update?

The evaluation also recorded whether a new CHESSCON execution was required. An execution was required when the update changed a condition that had to be verified through simulator output. An impact-only assessment was used when the evaluation concerned traceability and identification of affected areas rather than operational performance.

5.12 Relationship between measures and research questions

Table 5.12 connects each research question to its evidence. RQ1 uses the literature synthesis, RQ2 uses the implemented information structure, and RQ3 uses the controlled observations. This prevents simulation output from being used as the sole basis for all three answers.

Table 5.12: Research questions and evaluation measures.

Research question or result area	Evaluation evidence
RQ1	Focused literature synthesis and the baseline responsibilities derived in Chapter 2, using the method in Chapter 3
RQ2	Controlled records, mappings, trace structure, consistency checks, configuration guidance, change-impact structure and evidence rules
RQ3: modelling effort	Preparation, interaction and verification measurements
RQ3: traceability	Internal trace completeness
RQ3: consistency	Pass, managed-warning and failure rates
RQ3: scenario-change handling	Change-impact coverage and execution-check definition
Supporting correctness check	Indexed fleet count, completed-job rate, required identities and execution acceptance
Supporting auditability check	Evidence-completeness and data-integrity measures

Chapter 6

Results and Answers to the Research Questions

This chapter applies the measures defined in Chapter 5 and answers the three research questions. The chapter is organised by research question so that each answer remains directly connected to its supporting evidence.

The findings come from four sources: the focused literature review, the completed workflow-support package, the controlled CHESSCON executions and the three later scenario changes. Results taken from CHESSCON were calculated from the indexed result tables. Screenshots explain the interface but are not treated as the primary source of numerical evidence.

6.1 Answer to Research Question 1

> **RQ1: How can existing research be used to define the manual baseline for converting a scenario into a simulator configuration?**

Existing research was used to identify the responsibilities that must be performed when a written scenario is converted directly into a simulator configuration. The literature did not provide a ready-made baseline for CHESSCON. Instead, it described recurring responsibilities relating to interpretation, mapping, traceability, consistency, verification and provenance.

These findings were combined into a six-stage baseline workflow. The workflow begins with the written scenario and ends with preserved configuration and result evidence. It represents a normal direct process in which a person uses the available scenario documents, simulator guidance and product knowledge without the additional information structure provided by the proposed workflow.

6.1.1 Findings from the focused literature review

Six areas of recent software-engineering research contributed to the baseline definition. Table 6.1 shows how each area was converted into a responsibility within the baseline workflow.

Table 6.1: Derivation of the baseline workflow from existing research.

Literature area	Finding relevant to scenario conversion	Responsibility included in the baseline workflow
Semantic interpretation [35]	Meaning differs between representations.	Preserve values, units and context when interpreting the scenario.
Information mapping [23]	Source-to-target relationships require decisions.	Identify the relevant simulator object and setting.
Traceability [17]	Related documents do not automatically supply usable links.	Retain the source-to-setting relationships available during the work.
Configuration consistency [29]	Related representations require continuing agreement.	Check connected settings against the scenario.
Simulation verification [59]	Evidence must address stated properties.	Apply predefined conditions to the configuration and output.
Provenance [55]	Results require production context.	Preserve the connection between sources, configuration and result evidence.

The literature therefore provided the reasoning behind the baseline. It did not define the exact CHESSCON controls to use. Instead, it identified the activities that a credible direct conversion workflow must contain.

6.1.2 Baseline workflow used in this study

The exact baseline workflow established in Chapter 2 was used in the evaluation. It contained six stages: examine the scenario, interpret the information, map the information to the simulator, configure the simulator, verify the configuration and simulation result, and record the available evidence.

The established baseline appears in Figure 2.1. Traceability and consistency apply across stages. The empirical comparison used review, checking, execution and evidence activities around an existing model, not complete model construction.

The baseline was not created as an intentionally weak comparison. It allowed the written scenario, CHESSCON guidance and normal product knowledge to be used. What it did not include was the controlled information model, predefined mapping records, primary trace table, framework-audit check register and change-impact records supplied by the proposed workflow.

The result for RQ1 is therefore a defined comparison point grounded in the literature, with explicit stage responsibilities and cross-cutting checks. The synthesis establishes what belongs in the baseline; it does not establish how often the identified risks occur.

6.2 Answer to Research Question 2

> **RQ2: How should the proposed software workflow be designed to support this conversion while maintaining traceability and managing later scenario changes?**

The proposed workflow should place a controlled information layer between the written scenario and CHESSCON. This layer should store the scenario information

once, describe how that information maps to CHESSCON, connect every important value to its source and verification evidence, define consistency checks before execution, record the effect of later changes and preserve the evidence needed to review the result.

CHESSCON remains a separate external simulator. The proposed workflow does not operate it automatically. A person still opens the model, checks the settings, starts the simulation and reviews the output. The software contribution is the controlled and reviewable information structure used around those activities.

6.2.1 Structure present in the evaluated proposed workflow

The evaluated package contained the information types defined in Table 4.4: controlled records, mappings, primary traces, checks, guidance, change records and an evidence catalogue. Their presence is implementation evidence; it does not itself establish improved productivity or complete source capture.

The 30 controlled records were stored in YAML. YAML is a text format that represents named values in a readable structure. The mappings were also stored in YAML. The traceability table and evidence catalogue used comma-separated value files, which can be read as tables. The consistency and configuration material used Markdown, which is a plain-text document format.

These formats were selected because they are readable without specialised software and can be reviewed independently of CHESSCON. Table 6.2 connects each requirement from Chapter 2 to its implemented support and available evidence. This makes the distinction between a design mechanism and a measured benefit explicit.

Table 6.2: Workflow requirements, implementation and evidence.

Req.	Implemented responsibility	Evidence and assessment boundary
WR1	Shared controlled information	30 records; no measured reduction in duplicate values.
WR2	Record source, controlled content or value, unit where applicable, and meaning.	Recorded fields and traces; independent source-capture coverage unknown.
WR3	Facts separate from mappings	Separate scenario and mapping files; nine mapping rules.
WR4	Links to settings, checks and evidence	30 primary internal traces; not proof of all source requirements.
WR5	Predefined checking guidance	Fifteen framework-audit checks and eight common execution acceptance checks; no measured error-prevention effect.
WR6	Change-impact records	Three documented cases; workload and fleet executed, distance impact-only.
WR7	Identity and provenance	Project identities, hashes and result pairs retained; common reader uses status selection and index count.
WR8	Explicit human boundary	CHESSCON operation remains manual; no importer or executable interface demonstrated.

6.2.2 Separation of responsibilities

The proposed workflow separates information that performs different responsibilities. The controlled information records contain selected scenario requirements, simulator context and evaluation-evidence information. The mapping layer explains where that information belongs in CHESSCON. The traceability table records how the source, configuration responsibility, verification check and evidence are connected. The consistency layer checks whether related information agrees. The configuration guide explains what must be checked in CHESSCON. The evidence catalogue identifies the supporting files.

This separation avoids placing every responsibility in one large document. One controlled record can be referenced by mappings, checks and evidence records. This is a design mechanism intended to minimise duplication; duplicate values were not counted before and after the comparison.

6.2.3 Traceability result

All 30 controlled information records had at least one complete primary trace path. Substituting the retained counts into Equation 5.6 gives:

$$\frac{30}{30} \times 100 = 100\%. \tag{6.1}$$

This means that every selected controlled record had a documented connection to its main implementation or verification responsibility. It does not establish completeness against all source requirements: an independent source-capture inventory was not retained.

The baseline workflow had no equivalent consolidated traceability table available before execution. This does not mean that the baseline had no source information. Figure 6.1 therefore reports the proposed register's internal completeness without assigning an unsupported zero percentage to the baseline.

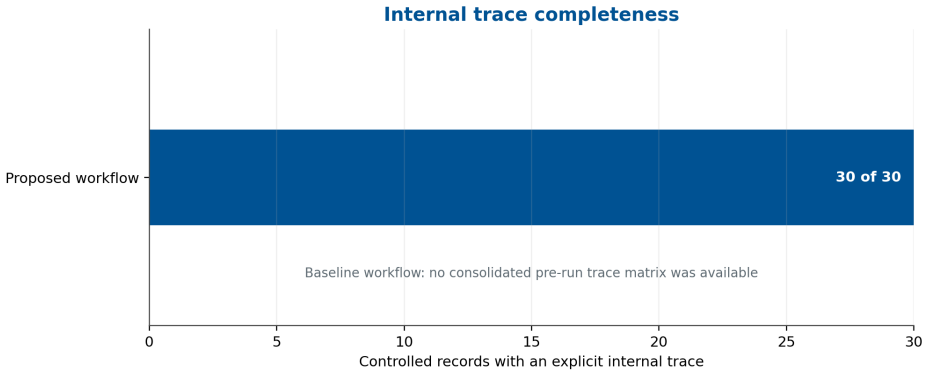


Figure 6.1: Internal trace completeness in the workflow-support package: 30 of 30 controlled records. A comparable baseline percentage is unavailable; absence of a consolidated table is not a measured zero for all baseline traceability.

6.2.4 Consistency structure

The fifteen framework-audit checks produced 11 passes, four managed warnings and no failures. Figure 6.2 separates those classes rather than combining warnings with passes. These are retained framework-audit outcomes, not an additional assessment of each accepted execution. A managed warning records a known interpretation issue and its effect; it is neither an unqualified pass nor a failure.

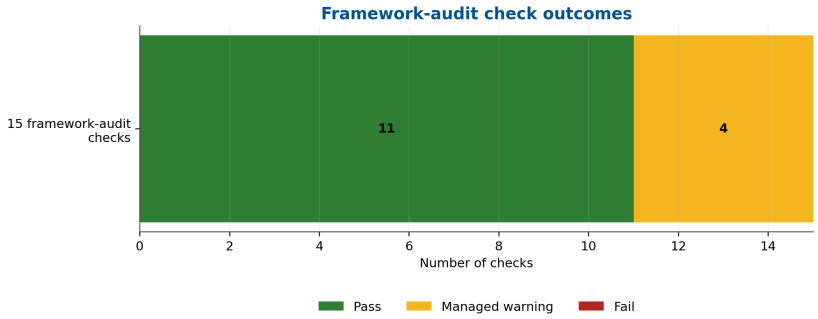


Figure 6.2: Outcome of the fifteen framework-audit checks: 11 passes, four managed warnings and no failures.

The four warnings are explained in Table 6.3.

Table 6.3: Managed warnings in the proposed workflow.

Warning	Recorded issue	Effect on the evaluation
Simulation period	Historical material mentioned 55 hours, while the evaluated model used 72 hours.	Report 72 hours and keep the historical material separate.
Movement distance	The scenario stated 50 metres, while CHESSCON movement results depend on the represented route geometry.	Do not compare the stated distance directly with total simulated travel.
Historical energy fields	Earlier material contained activity-energy values that could be mistaken for direct fuel measurements.	Do not make a fuel-performance conclusion from those fields.
Historical small test	Older evidence contained a test with 200 jobs and two forklifts.	Exclude the test from the accepted reference-scenario evidence.

The important result is not simply that eleven checks passed. The proposed workflow made the four unresolved interpretation issues visible before the result was reported.

6.2.5 Support for later scenario changes

The proposed workflow records changes by identifying the controlled information that changes and the related mappings, CHESSCON areas, checks and evidence that may need to be reviewed. Three changes were examined, as shown in Table 6.4.

Table 6.4: Status of the three examined scenario changes.

Scenario change	Changed condition	Evaluation status
Increase in movement jobs	Total jobs increased from 2,000 to 2,400.	Executed; total jobs and transfer counts and targets verified
Reduction in forklifts	Active forklifts reduced from four to three.	Executed; three active forklifts and 2,000 completed jobs verified
Increase in stated distance	Stated movement distance increased from 50 metres to 75 metres.	Impact analysis only; no simulation result claimed

The distance change demonstrates the boundary of the evidence. The proposed workflow identified what would be affected, but the model was not executed for this change. Therefore, no claim is made about its effect on time, energy use, utilisation or completed work.

Together, the records, relationships and explicit human boundary answer RQ2 with an implemented workflow design. The result is the organisation supplied to the work; its observed comparison with the baseline is addressed next.

6.3 Answer to Research Question 3

6.3.1 Execution environment

The author identified an Acer Swift SFG14-63 with an AMD Ryzen 5 8645HS processor (6 cores, 12 logical processors), 16 GB of memory and a 512 GB Samsung SSD as the trial computer. These details were supplied retrospectively and were not preserved in the execution record. Windows 11 Home, 64-bit, build 26200 was also reported retrospectively; the exact execution-time Windows build has not been established. The historical CHESSCON version remains unavailable, as explained in Section 5.8.1. These reported details therefore supplement the environment description without establishing a complete historical configuration. Additional resources cannot change the results already recorded, but may affect runtime on a later run. Output equivalence across computers was not tested.

> **RQ3: How does the proposed workflow compare with the manual baseline under the same scenario and evaluation conditions?**

Both workflows produced results satisfying the same eight execution acceptance checks. Each accepted result contained four active battery-electric forklifts, 2,000 completed jobs, the required ship-side transfer object, the required incoming and outgoing workload and two charging stations.

The workflow-support package contained explicit traces, predefined checks, an evidence catalogue and change-impact records not supplied as equivalent consolidated baseline registers. The available timing observations were not sufficient to demonstrate a reliable reduction in modelling effort.

6.3.2 Agreement on the eight acceptance conditions

Eight acceptance conditions were applied to both executions. Table 6.5 presents the observed results.

Table 6.5: Acceptance results for the two workflow executions.

Acceptance condition	Baseline workflow	Proposed workflow	Result
1. Active forklifts	4	4	Pass
2. Forklift type	`FL-1` in all active rows	`FL-1` in all active rows	Pass
3. Completed workload	2,000	2,000	Pass
4. Active ship-side transfer rows	1	1	Pass
5. Ship-side transfer identity	`QC-1`	`QC-1`	Pass
6. Transfer jobs and targets	Incoming, outgoing and both targets: 1,000 each	Incoming, outgoing and both targets: 1,000 each	Pass
7. Active charging rows	2	2	Pass
8. Charging identities	`CHARGER-1` and `CHARGER-2`	`CHARGER-1` and `CHARGER-2`	Pass
Overall result	**8 of 8 passed**	**8 of 8 passed**	**Accepted**

No simulation error was reported for either accepted execution. Figure 6.3 compares the common target with the observed total in each run, making the limited equivalence claim visible.

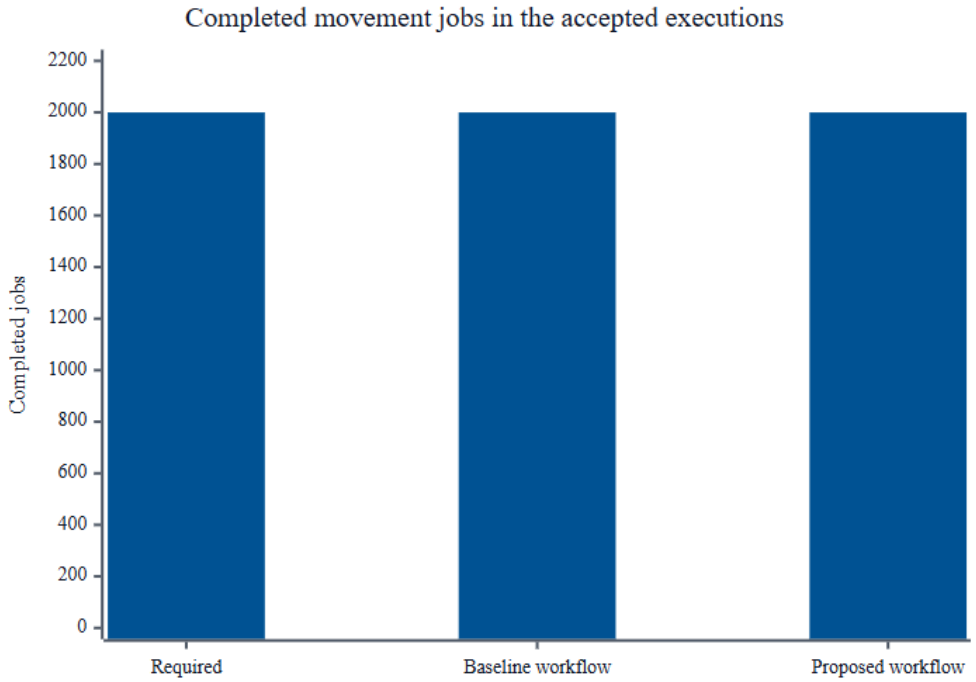


Figure 6.3: Required and observed completed-job totals for the baseline and proposed workflow executions.

Together, the accepted results establish equivalence on the eight selected acceptance conditions. They do not show improved operational performance. Table 6.6 identifies which defined conditions have accepted-run evidence and which rely only

on configuration records or illustrations, preventing the eight-check result from being extended to the whole scenario.

Table 6.6: Scenario coverage and the limits of accepted-run evidence.

Defined condition	Recorded representation	Accepted-run evidence
Four electric forklifts	Fleet records and equipment-type illustration	Four active rows, all type `FL-1`; battery behaviour not separately accepted.
2,000 jobs, equally incoming and outgoing	Workload records and configuration illustration	Total 2,000; incoming, outgoing and targets 1,000 each.
Ship-side transfer point	Required identity `QC-1`	One active row with that identity.
Two charging stations	Named charging objects	Two active rows with the required names; charging policy not verified by these checks.
72 hours and one run	Configuration and execution records	Documented settings; not separate conditions in the eight output checks.
Additional CHESSCON equipment-input/context values	Controlled values and Input illustration	Not individually tested by the accepted-run checks.
Battery capacity and charge thresholds	Scenario records; capacity visible in the supplied Input illustration	No independent accepted-run test of capacity or thresholds.
Item type, mass, storage destination and distance	Scenario definition and layout illustration	No complete item-level, destination or route-distance acceptance test.

Screenshots illustrate configuration, not the identity of an accepted run unless that provenance is recorded. Equal output totals establish the eight-check equivalence only; they cannot establish that every requirement above was preserved.

6.3.3 Distribution of work across the forklifts

The completed jobs were distributed slightly differently among the four forklifts. The labels `FL-001`, `FL-002`, `FL-01` and `FL-02` are the CHESSCON names assigned to the four individual forklift units. Table 6.7 gives the exact indexed values.

Table 6.7: Completed jobs by active forklift.

Forklift	Baseline workflow	Proposed workflow	Difference
`FL-001`	504	503	-1
`FL-002`	506	505	-1
`FL-01`	550	550	0
`FL-02`	440	442	+2
Total	**2,000**	**2,000**	**0**

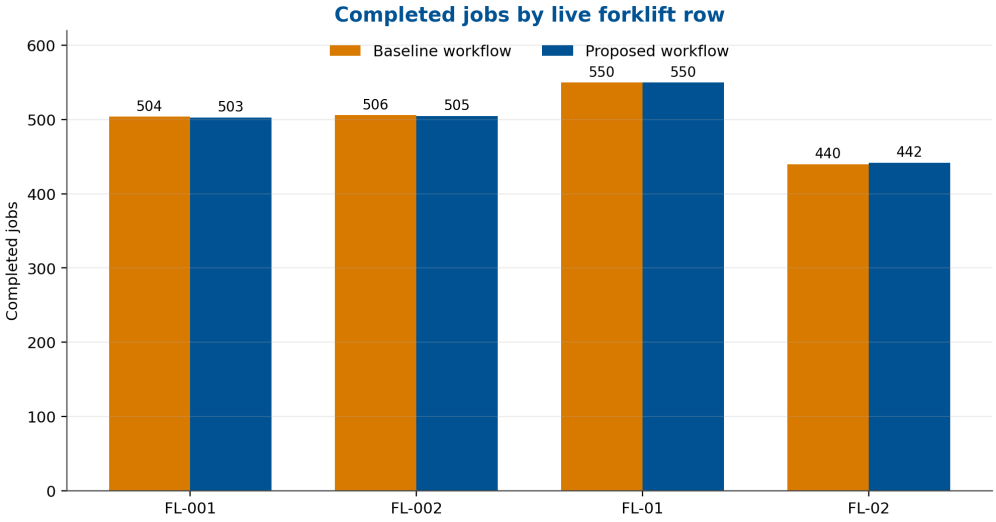


Figure 6.4: Per-forklift allocation in the two accepted executions. The small row-level differences do not change the aggregate total.

Figure 6.4 shows where the small allocation differences occur. No fixed random seed was documented: a seed is a value used to reproduce the same random sequence in repeated simulations. Exact row-level equality was therefore not required. This does not establish the cause of each difference; it explains the use of aggregate acceptance conditions.

The largest row-level difference was two completed jobs. The total remained 2,000 in both executions. The row-level differences do not contradict equivalence on the eight selected acceptance conditions, which do not require identical allocation of work to each forklift.

6.3.4 Traceability comparison

Table 6.8 contrasts the available trace structures. The baseline used source information and guidance without a consolidated trace table. The proposed workflow supplied a primary trace for every selected controlled record. The comparison describes these structures, not a measured difference in understanding.

Table 6.8: Traceability comparison between the workflows.

Measure	Baseline workflow	Proposed workflow
Controlled information records	Not stored as a consolidated record set	30
Explicit primary trace links available before execution	No equivalent consolidated trace table	30
Controlled records with a primary trace	Not applicable	30 of 30
Internal trace completeness	Not calculated from an equivalent baseline table	100%
Mapping rules	No separate mapping file	9

The absence of a consolidated baseline trace table must not be interpreted as zero knowledge, zero documentation or zero available evidence. The workflow-support package supplied a complete internal trace register. A reviewer with authorised access to the complete private register could follow its selected records through the documented relationships. The PDF presents the structure and selected examples; it does not allow independent checking of every register entry.

6.3.5 Consistency comparison

The baseline did not include an equivalent consolidated consistency register, although settings could still be checked. Table 6.9 reports the proposed register's class counts and percentages using Equations 5.7–5.9. It is a description of that audit, not a paired error-rate comparison.

Table 6.9: Outcome of the proposed workflow's framework-audit checks.

Check result	Number	Percentage
Pass	11	73.33%
Managed warning	4	26.67%
Fail	0	0%
Total	**15**	**100%**

This result records explicit checking guidance and its audit outcomes. It does not prove that the proposed execution contained fewer real inconsistencies than the baseline execution because the baseline did not provide an equivalent predefined check set for a direct numerical comparison.

6.3.6 Configuration identity and setup incidents

Two attempted starts during baseline controller setup were rejected after CHESSCON reported a connection message containing `QC-03`, an unexpected identifier for a ship-side transfer object. The intended ship-side transfer object was `QC-1`. These were controller-troubleshooting incidents, not observed scenario-mapping mistakes or accepted-run simulation failures. They did not change the accepted result files. The separate proposed setup recorded no rejected start; the events are not used to infer error prevention.

Table 6.10: Accepted execution and configuration events.

Event	Baseline workflow	Proposed workflow	Interpretation
Accepted executions	1	1	Both produced accepted results
Model edits during the accepted comparison	0	0	The comparison concerned workflow support around an existing model
Accepted-run simulation errors	0	0	No accepted execution error was recorded

Table 6.10 reports accepted executions and model edits. The rejected setup events are described separately above and do not establish that the proposed workflow prevented errors. Both accepted executions satisfied the common conditions.

6.3.7 Preparation-time observations

Figure 6.5 compares the recorded preparation intervals: 71.60 seconds for the baseline workflow and 27.19 seconds for the proposed workflow. Each bar represents one observation, not an average across repeated trials.

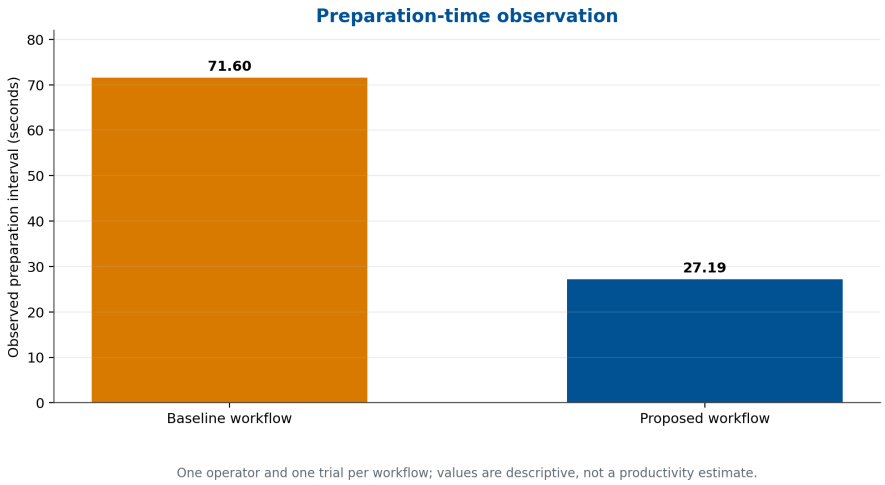


Figure 6.5: Single observed preparation interval for each workflow.

These are single descriptive pilot observations. They do not establish a reduction in total modelling effort, improved productivity or cost-effectiveness.

The event record retains preparation durations of 71.5967 seconds for the baseline workflow and 27.1861 seconds for the proposed workflow, reported here as 71.60 and 27.19 seconds. Controller diagnostics and user waiting were excluded from recorded workflow timing. The two rejected baseline starts are separate setup events, not components assigned to the 71.60-second interval. The retained event records preserve the reported preparation durations and trial order. However, the detailed activity logs are insufficient to establish that both intervals covered the same preparation activities or fully matched the intended boundaries in Chapter 5. One observation per workflow, asymmetric setup work and possible learning effects limit interpretation. Table 6.11 reports the status of every declared measure, including those that could not be calculated.

Table 6.11: Reporting status of all declared evaluation measures.

Measure	Supported observation or reason unavailable
Completed jobs and completion rate	Both reference runs: 2,000 and 100%; workload change: 2,400 and 100%; fleet change: 2,000 and 100%.
Active counts and identities	Both reference runs pass eight checks; changed fleet has three active forklift rows.

Table 6.11 (continued)

Measure	Supported observation or reason unavailable
Preparation observations	Baseline: 71.60 seconds; proposed: 27.19 seconds. One pilot observation per workflow; not a measure of total effort or a causal performance effect.
Interaction, simulation, verification, evidence handling and total time	Complete matching activity boundaries unavailable. Output-write windows of 5.003 and 4.416 seconds are not simulator runtimes. Write-to-acceptance lags of 117.077 and 11.981 seconds are not verification effort.
Setup incidents and configuration corrections	Two starts were rejected during baseline controller troubleshooting; none was recorded in the separate proposed setup. These are setup incidents, not comparative workflow-error results. Zero model edits were recorded for accepted trials; the total setup-correction count is unavailable.
Repeated lookups, simulator areas and evidence-processing actions	Not instrumented sufficiently for complete comparable counts; not estimated.
Internal trace completeness and mapping count	Proposed: 30/30 (100%) and nine mappings. No equivalent baseline register; independent source-capture percentage unavailable.
Framework-audit rates	Proposed retained audit: 11/15 (73.33%) pass, 4/15 (26.67%) managed warning, 0/15 fail; not paired error rates.
Physical, active and inactive forklift rows	Reference runs: 4, 4, 0 each. Reduced-fleet case: 4, 3, 1.
Evidence completeness	Percentage unavailable: no frozen required-item denominator. Inventory: 37 evidence-catalogue entries.
Five-category change coverage	Retrospective document assessment below; not execution-time effort or a paired comparison.

6.3.8 Results of the scenario changes

The three changes examined different parts of the scenario and different forms of change evidence. Table 6.12 reports the observed results and their supported interpretation.

Table 6.12: Results of the later scenario changes.

Change	Preserved result	Supported finding
Increase in movement jobs	Four active forklifts completed 641, 607, 605 and 547 jobs, totalling 2,400.	The changed workload was executed and satisfied the new aggregate requirement.
Reduction in forklifts	Three active forklifts completed 667, 728 and 605 jobs, totalling 2,000.	The live result represented the changed fleet and completed the required total.
Increase in stated distance	Affected scenario information, mappings and CHESSCON areas were documented.	The impact could be followed, but no simulated performance result was produced.

For the increased workload, the four active forklift rows completed:

$$641 + 607 + 605 + 547 = 2{,}400 \text{ jobs.} \tag{6.2}$$

The preserved ship-side transfer record also supports the changed workload in both directions. A read-only recheck of the table/index pair found one active `QC-1` record, the ship-side transfer object, with 1,200 incoming jobs, 1,200 outgoing jobs, an incoming target of 1,200 and an outgoing target of 1,200. Its selected-record count agreed with the companion index. These four fields strengthen the aggregate completion check without extending it to every scenario condition; they were rechecked from preserved output, not from a new simulation.

For the reduced fleet, the three active forklift rows completed:

$$667 + 728 + 605 = 2{,}000 \text{ jobs.} \tag{6.3}$$

Table 6.13: Retrospective coverage of five change-impact categories.

Recorded category	Workload	Fleet	Distance
Changed information	Explicit	Explicit	Explicit
Affected mappings	Explicit	Explicit	Explicit
CHESSCON areas	Explicit	Explicit	Explicit
Checks to repeat or update	2,400-job output check	Three-row and job checks	General impact review; no specific repeated check
Required evidence	Workload and output records	Deployment and output records	Layout and impact evidence
Explicit categories	5/5 (100%)	5/5 (100%)	4/5 (80%)

Table 6.13 applies the five-category rule retrospectively to the preserved change-impact report. A named check is required, not general review guidance. The distance record does not name a specific repeated check, so that category is not counted. This does not make the impact-only case a failed simulation. No corresponding baseline scores were recorded, and these percentages do not measure speed or ease of change.

6.3.9 Indexed-data integrity in the reduced-fleet result

The physical forklift result table retained four records after the fleet was reduced. However, only three table records had the active status, and this count agreed with the companion index. The remaining `FL-002` record, named after one individual forklift unit in CHESSCON, was stale and was not part of the accepted result.

Figure 6.6 separates stored rows from selected rows. Table 6.14 gives their counts and identities. The reader used table statuses and an index-count cross-check, not complete index traversal; Section 5.8.1 explains that limitation.

Table 6.14: Physical and active views of the reduced-fleet result.

Evidence view	Row count	Correct interpretation
Physical forklift table	4	Contains three active records and one retained stale record
Companion index	3	Supplies the expected active count, not the reader's record identities

Table 6.14 (continued)

Evidence view	Row count	Correct interpretation
Parsed accepted summary	3	`FL-001`, `FL-01` and `FL-02`; total 2,000 jobs
Excluded physical row	1	`FL-002`, the CHESSCON name of the removed forklift; inactive

If all four physical rows had been added without status-based selection and the index-count cross-check, the result would have been interpreted incorrectly. This finding shows that record-selection rules are important parts of simulator verification.

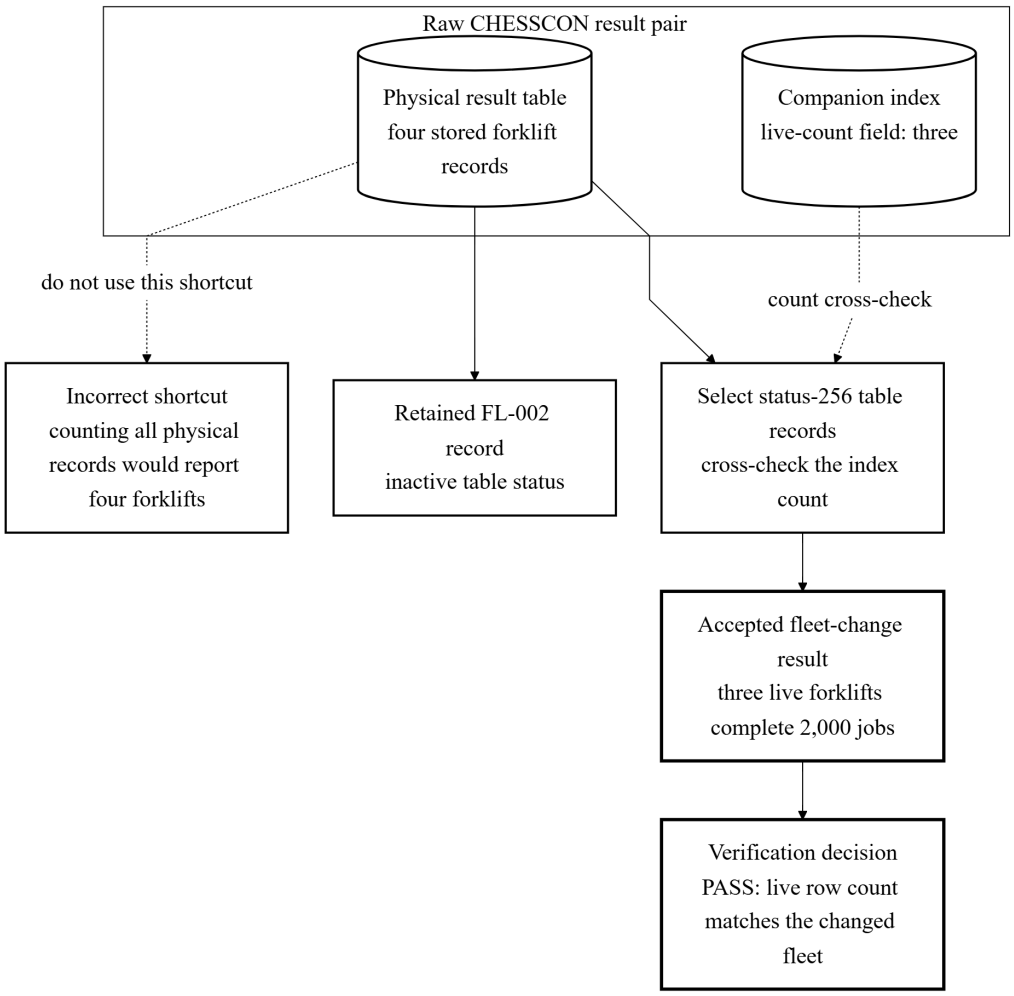


Figure 6.6: The physical result table retained four records, but status-based selection and an index-count cross-check retained three records for the accepted result.

6.3.10 Comparison summary and result for RQ3

The workflow-support package provided explicit internal traces, predefined checks, an evidence catalogue and change-impact records, while both workflows satisfied the same eight output checks. The evaluation did not establish a reliable reduction in modelling effort or an improvement in simulator performance.

Table 6.15 combines the findings at the level supported by the available evidence.

Table 6.15: Overall comparison of the baseline and proposed workflows.

Evaluation area	Baseline workflow	Proposed workflow	Supported conclusion
Functional correctness	Passed all eight acceptance conditions	Passed all eight acceptance conditions	Equivalent on the eight selected output checks
Completed jobs	2,000	2,000	No difference in the required aggregate result
Explicit traceability	No consolidated pre-run trace table	30 of 30 controlled records traced	Complete internal trace register; source-capture coverage unknown
Framework-audit support	No equivalent consolidated check package	15 framework-audit checks: 11 passes, four managed warnings and zero failures	Checks and known concerns were visible in the proposed workflow
Evidence organisation	Evidence available across source and result material	37 evidence-catalogue entries	An explicit catalogue was supplied; lookup effort was not measured
Change-impact information	No separate impact-record structure	Five categories documented for the workload and fleet changes; four for the distance change, which named no specific repeated check	Affected categories were documented; ease was not measured
Preparation observation	71.60 seconds	27.19 seconds	The proposed observation was shorter, but no general effort reduction was established
Accepted simulation errors	0	0	Both accepted executions completed without an observed error
Simulator performance improvement	Not evaluated	Not evaluated	No claim of improved simulation or terminal performance

Under the same acceptance conditions, both workflows satisfied the eight selected output checks. Each accepted execution represented four active forklifts completing 2,000 jobs, the required incoming and outgoing workload and two charging stations.

The workflow-support package supplied observable structures: explicit traces for all 30 controlled records, nine mappings, fifteen framework-audit checks and 37 evidence-catalogue entries. The catalogue is retained with the study documentation and identifies the purpose and location of supporting material. Results and supporting evidence were presented to the university supervisor during supervision. This does not constitute independent replication. The proposed workflow also documented which information and simulator areas were affected by the three scenario changes.

The proposed preparation interval was shorter in the recorded comparison. Because the evidence contains one observation from each independent setup and the setups recorded different preparation events, the times are reported as descriptive observations rather than a general estimate of modelling effort.

Chapter 7

Discussion

7.1 Overall interpretation of the findings

The evaluation examined two workflows for reviewing, checking, executing and documenting an already configured CHESSCON model. The baseline workflow relied on direct interpretation of the scenario and its supporting information. The proposed workflow organised the same information through controlled records, mappings, trace links, consistency checks, configuration instructions and evidence references.

Both workflows satisfied the same eight output checks. This establishes agreement on those conditions, not preservation of every scenario requirement. The main difference was therefore not the operational behaviour of the simulated system. It was the presence of explicit information structures supporting configuration review and change analysis; understanding and usability were not independently measured.

The baseline workflow depended more strongly on knowledge held by the modeller. The proposed workflow moved part of this knowledge into visible and reviewable records. Within the retained private package, these links record why a value was included, where it was configured, which check applied and what evidence was retained. Inspecting all such links requires authorised access; the thesis illustrates selected relationships.

Table 7.1 connects the observed findings with their broader software-engineering meaning.

Table 7.1: Interpretation of the main evaluation findings.

Finding	Meaning in this study	Software-engineering relevance
Both accepted executions satisfied the same conditions	The selected output conditions agreed in both accepted runs.	Explicit information structures accompanied an equivalent selected output result.
Scenario information was held in one controlled structure	Selected values had a common recorded location.	Provides references to shared records; duplication reduction was not measured.
Mappings and trace links were recorded explicitly	The relationship between scenario information and CHESSCON responsibilities became visible.	Supports review, maintenance and later verification.
Consistency checks were defined before result interpretation	Expected agreements and known uncertainties were documented.	Provides explicit check definitions and recorded managed warnings.

Table 7.1 (continued)

Finding	Meaning in this study	Software-engineering relevance
Change-impact records identified affected information	Later updates could be followed through their dependencies.	Supports controlled maintenance of the configuration.
Evidence was catalogued by purpose and location	A reviewer could locate the material supporting a result or decision.	Provides a catalogue for provenance review; retrieval effort was not measured.
Result tables were interpreted with their indexes	Records were selected by table status and their count checked against the index.	Excluded the retained stale record from the reported active results in this case.

7.2 From personal knowledge to recorded knowledge

The main contribution of the proposed workflow is that important modelling knowledge is made explicit. In the baseline workflow, a modeller reads the scenario, identifies the corresponding CHESSCON concepts and verifies the resulting configuration. This process can work successfully, as demonstrated by the accepted baseline execution. However, much of the reasoning can remain in the modeller's memory or be distributed across documents and simulator screens.

Knowing that four electric forklifts are required, for example, is only one part of the configuration task. The modeller must also know which CHESSCON equipment type represents those forklifts and where the four instances are deployed. The result records must confirm the four active units in the accepted execution. These connections may be clear to the person performing the work, but they are not automatically preserved as one reviewable relationship.

The proposed workflow records these connections outside CHESSCON. It does not replace professional judgement. Instead, it preserves the result of that judgement so that it can be inspected later. This distinction is important because a simulation model can execute correctly while its configuration reasoning remains difficult to recover. Successful execution shows that the simulator accepted the configuration; it does not explain why every setting was selected or which scenario statement it represents.

The evidence catalogue supports this recorded knowledge through 37 evidence-catalogue entries in the retained study documentation, including references to source material, configuration evidence and interface images. Its existence establishes an inventory, not complete evidence coverage. Appendix A presents supplementary interface and result illustrations; access to original licensed project files remains restricted.

7.3 Separation of responsibilities

Table 7.2 shows how the proposed workflow assigns related responsibilities to named records. The separation permits a reviewer to distinguish a required value, its mapping and its verification evidence. Its effect on user understanding was not measured.

Table 7.2: Responsibilities separated in the proposed workflow.

Responsibility	Location in the workflow	Reason for separation
Organise selected conditions, context and evidence	Controlled information model	Keeps their meaning explicit outside the CHESSCON interface.
Explain how information is represented	Mapping rules	Separates interpretation from the original value.
Connect sources, settings and evidence	Traceability table	Makes the conversion reviewable.
Define what must agree	Consistency checks	Separates verification rules from the final result.
Identify the effect of an update	Change-impact records	Makes related impacts explicit when a changed value is reviewed.
Explain simulator actions	Configuration guide	Supports consistent use of CHESSCON.
Identify supporting material	Evidence catalogue	Gives retained evidence a stable identity and purpose.
Perform simulator actions	Human-operated CHESSCON boundary	Makes responsibility for simulator actions explicit without claiming an automatic importer.

The scenario states what is required. A mapping explains how that requirement is represented. A check defines how the representation is verified. Evidence shows what was inspected or produced. Keeping these responsibilities separate provides a way to refer to shared values instead of maintaining competing definitions. Actual duplication reduction was not measured, and the package does not enforce updates automatically. Its demonstrated property is the presence of named relationships that can be inspected when a value changes.

This design follows recent work that treats shared meaning, transformation, traceability and provenance as related but distinct software concerns [2, 14, 17, 35, 54].

7.4 Configuration identity and execution state

The evaluation showed that correct values are not sufficient if the active project or simulator object is uncertain. Configuration identity means knowing exactly which project, object and result set are being examined.

The setup incidents reported in Section 6.3.6 illustrate the need for explicit stop rules and project-identity checks. A familiar project name is insufficient when several similar project copies exist. The complete project path, expected object identifiers and accepted-result location must agree before an execution is treated as valid.

The proposed setup recorded no rejected start. This is the observed outcome of that separate setup. Project copies kept the files separate, but did not remove possible learning by the same researcher. The two-versus-zero event count cannot establish that the proposed workflow caused fewer setup problems.

Configuration identity is a software-management concern rather than a terminal-performance concern. The event evidence makes the problem concrete: correct values must also belong to the intended project state. Computational reproducibility work similarly requires the relevant execution environment to be

identified [14]. This connection concerns the need for context, not a claim that the cited study observed the same CHESSCON controller problem.

7.5 Indexed-result integrity

The reduced-fleet scenario exposed an important result-handling issue. Its physical CHESSCON result table retained four forklift records even though only three forklifts were active. The reader selected three status-256 table records and checked their count against the index. It excluded the old `FL-002` record, a stored row named after one individual forklift unit.

This means that a physical row in a result file cannot automatically be treated as part of the current result. The table and its index must be interpreted together. The situation is similar to a database containing historical rows while a query selects only the currently active records. Counting every stored row would produce an incorrect interpretation even though the stored data had not been corrupted.

The proposed workflow addresses this risk through evidence rules that require the result table and its companion index to be preserved together. Parsed summaries improve readability, but they do not replace the original table-and-index pair. This supports the wider reproducibility principle that stored output must retain enough technical context to be interpreted correctly [20, 21].

7.6 Interpretation of the preparation observations

The preparation observations reported in Section 6.3.7 cannot isolate the effect of the workflow-support package. The baseline trial preceded the proposed trial, allowing learning to influence the later observation. The activity logs also do not establish that both recorded intervals covered equivalent work. These conditions prevent the observed difference from being interpreted as a reduction in overall modelling effort.

The proposed package was already constructed when its preparation interval began. Construction effort was not recorded, and maintenance effort was not measured. A practical efficiency assessment would need to include these costs alongside repeated measurements of preparation, configuration, verification and evidence handling.

7.7 Relationship to previous research

Liepert et al. [35] distinguish several forms of interoperability difficulty. The present case illustrates a narrow part of that problem: scenario meaning and simulator representation require an explicit connection even when a person performs the transfer. It does not evaluate exchange among multiple organisations or demonstrate a general interoperability solution. The comparison contributes a concrete information structure at one commercial-simulator boundary.

Fucci et al. [17] show that missing relationships can obstruct an intended industrial analysis. The present register takes the preventive design step of recording selected relationships, but its evaluation does not show prevention of future analysis failures. Ruiz et al. [54] also make the cost and purpose of industrial traceability important.

This qualifies the interpretation here: internal completeness is established, whereas long-term maintenance effort and benefit to a new user remain unmeasured.

Finally, the reproducibility-debt synthesis in [21] explains why missing execution context can create later work. Preserved table-and-index pairs and the documented selection rule address a specific part of that concern. Access restrictions and the lack of independent application-level verification remain. The study therefore contributes a bounded example of applying these established ideas together, rather than a replacement for the more general methods reviewed in Chapter 2.

The capability distinction is concrete. TracIMo [42] offers a methodology for introducing traceability around stakeholder goals. This workflow connects selected scenario facts, CHESSCON mappings, human checks and result evidence at one simulator boundary. Incremental transformation tools [44] execute model updates where supported interfaces and compatible models exist. This workflow instead records the affected information for an operator to review through the simulator's normal interface. That makes it a possible documentation layer in a licensed setting, not a superior or automated alternative.

7.8 Practical implications

The proposed workflow demonstrates how structured software documentation can support a commercial simulator without modifying the simulator itself. A team applying the approach would create one controlled information model, document each important mapping and define checks before execution. It would also record the expected effect of later changes and preserve evidence with stable identifiers. CHESSCON would continue to be operated through its normal interface.

This approach may be useful when direct system integration is unavailable or when project files should not be edited outside the supported application. Its intended benefit is structured review while respecting the simulator boundary. This study did not measure improved understanding or usability.

The implemented records and mappings could also provide a foundation for a future executable adapter. They already describe much of the information that such an adapter would require. The workflow evaluated in this thesis, however, remains transparent and human-operated.

Adoption also creates a maintenance responsibility. A recorded mapping is useful only while it still describes the relevant simulator version and scenario. If configuration changes are made without updating the connected records, the package can become a second source of outdated information. The practical implication is therefore not simply to add more documents. It is to assign responsibility for reviewing related records when a change is made and to retain evidence of that review. The current study establishes that such relationships can be recorded. It does not measure whether a team will maintain them over time or whether maintenance costs are lower than the work they replace.

The wider consequences concern the use of simulation evidence in decisions. An explicit record of what was checked can help distinguish a verified condition from an assumption. It must not make a narrow test appear more comprehensive than it is. A decision-maker who sees an accepted execution may otherwise infer that safety,

capacity or energy use has also been validated. Those properties were not established by the eight execution acceptance checks. The useful safeguard is the stated boundary of the evidence, rather than confidence based solely on a successfully completed run.

The study also has a sustainability boundary. Electric forklifts form the case setting, but the evaluation does not compare energy consumption, emissions or environmental outcomes between workflows. Maintaining structured records requires human work and digital storage; reviewing changes may also lead to additional simulation executions. Conversely, reusing an accurate record might avoid repeated investigation. These are possible consequences to examine in future use, not measured savings. Environmental or resource-efficiency claims would require separate measurements of the relevant work, computing resources and operational behaviour.

7.9 Threats to validity

The comparison used one scenario, one researcher and one accepted execution per workflow. The recorded outcomes show what happened in this case, but cannot establish a typical effect across people or projects. The separate project copies protected file state; they did not make the researcher independent between trials.

The preserved timestamps place the baseline before the proposed trial. Knowledge gained in the first trial may therefore have influenced preparation or simulator use in the second. The baseline also required controller troubleshooting that the proposed setup did not. Those diagnostics were excluded from recorded workflow timing, but the experience may still have affected later work. Neither the timing difference nor the rejected-start counts isolate a causal workflow effect.

Measurement was limited by incomplete activity logs and missing matching boundaries for several time categories. The reported preparation intervals cannot be treated as complete effort measurements. The proposed-workflow preparation observation concerns use of an already prepared workflow-support package and does not include the one-time effort required to design and populate that package. The fifteen framework-audit checks are distinct from the eight execution acceptance checks applied to each accepted run. The baseline lacked an equivalent predefined register, so the audit does not support a paired numerical claim of fewer inconsistencies.

Both workflows began around an already configured model. The study therefore does not measure construction from an empty project. CHESSCON-specific objects and index behaviour also limit transfer to other simulators. No fixed random seed was documented, so aggregate acceptance checks provide the comparison boundary; small differences between individual forklift records cannot be attributed to the workflow. The workload and fleet changes have executed evidence, while the distance change has impact-analysis evidence only, and the changes do not form repeated paired trials.

The designer also conducted the evaluation and was familiar with the package. The selected scenario may fit the design particularly well; independent scenario selection was not tested. The 30-record inventory does not establish complete capture of the source requirements, and the eight acceptance checks leave other scenario conditions untested. Independent project copies do not remove these sources of bias.

The two execution directories also differed in controller metadata even though the underlying model identity was checked. Those differences are part of the retained

provenance, not evidence that all directory contents were byte-identical. Restricted access to licensed models and raw tables limits independent replication. The thesis presents derived evidence, not all inputs needed to repeat the execution. The reader's status-based selection with an index-count cross-check is another boundary; it is not a full independent validation of CHESSCON's index semantics.

The historical search queries, screening decisions and exclusion process were not retained in sufficient detail for independent reproduction. The focused literature corpus is therefore purposive, and literature-selection bias may have influenced the responsibilities derived for RQ1. The source-specific synthesis makes the adopted responsibilities inspectable, but does not establish exhaustive coverage of the relevant research.

These limits support a descriptive conclusion about observable records rather than measured usability. Repeated, counterbalanced trials with complete timing would be needed to estimate productivity or error-prevention effects. Here, counterbalancing means varying workflow order across trials.

Chapter 8

Conclusions and Future Work

8.1 Conclusions

This thesis addressed an information problem that remains important even when simulation software is available and the model can run. A written scenario, a simulator setting and an execution result describe related aspects of the same task, but they do not automatically explain their relationship to one another. The research developed one contribution: a proposed workflow that makes those relationships explicit while retaining human operation of CHESSCON.

The workflow is implemented through a human-readable support package. Controlled information records state selected content and its meaning, with values and units where applicable. Mappings relate those records to simulator responsibilities. Trace links connect the information to checks and evidence, while change-impact records identify the information that needs attention after an update. A configuration guide brings those records into the human task of inspecting and using CHESSCON. The package is therefore an organised means of supporting work, not an executable importer or a replacement for the simulator.

The focused literature review established the baseline against which this contribution was examined. It brought together interpretation, mapping, traceability, consistency, verification and provenance as connected responsibilities. This answers the first research question through a reasoned synthesis rather than by assuming that a ready-made CHESSCON workflow existed in a single publication. The resulting baseline remains a credible direct process: it allows ordinary source documents, simulator guidance, notes and professional judgement. Its distinction from the proposed workflow is the absence of the additional shared package, not the absence of careful work.

The second research question is answered by the implemented organisation of information. The workflow separates what a scenario states from how CHESSCON represents it and from the evidence used to check it. This separation is important because those responsibilities can change independently. A scenario value can change without changing the name of a simulator field, while a simulator representation can change without changing the intended scenario meaning. Recording the connection makes these differences inspectable. It does not guarantee that every dependent record will be updated correctly; that remains a human verification responsibility in the evaluated implementation.

For the third research question, the controlled comparison provides a bounded result. Both accepted executions satisfied the same eight selected output conditions.

The proposed workflow had explicit internal traces, mappings, a retained check register and documented change impacts. The comparison did not establish a causal reduction in effort or errors. Table 8.1 brings together these findings without treating a document count, a completed simulation and a timing observation as equivalent forms of evidence.

Table 8.1: Contribution established by each part of the study.

Part of the study	What was established	Interpretation boundary
Literature synthesis	A direct workflow with connected information and checking responsibilities.	Not an exhaustive survey or a measured failure rate for manual work.
Workflow implementation	Explicit records, mappings, traces, checks, instructions and evidence responsibilities.	Not automatic transfer into CHESSCON.
Reference comparison	Agreement on the eight accepted output conditions.	Not verification of every scenario property or better physical operation.
Change evidence	Executed workload and fleet changes, with a separate distance impact analysis.	Not repeated paired change trials under both workflows.
Effort observations	Recorded preparation intervals and setup-event counts.	Not a demonstrated productivity or error-prevention effect.

The traceability result covers the selected controlled records only; it does not establish complete capture of the original scenario. The consistency register describes recorded checks and managed uncertainty, not a paired measure of execution errors. In the reduced-fleet case, result interpretation required status-based row selection and an index-count cross-check. That procedure supports the reported totals but is not an independent validation of all CHESSCON file semantics.

The single case, already configured model, fixed trial order and incomplete activity logs limit interpretation and transfer, as discussed in Chapter 7. These conditions bound the conclusions; they do not invalidate the preserved observations.

The overall conclusion is that a human-operated simulation workflow can be supported by explicit, connected information without modifying the commercial simulator. In this case, the records made the intended links and verification responsibilities available for inspection alongside equivalent selected execution results. Whether that support reduces total work, improves understanding or transfers successfully to other settings requires further evaluation.

8.2 Future work

Further work should build on the implemented relationships while testing the activities not established by this study. The next step is not simply to add more records. It is to make each proposed extension answer a specific unresolved question with evidence suited to that question. Table 8.2 orders these extensions by the uncertainty they address.

Table 8.2: Priorities for further development and evaluation.

Next study	Required work	Evidence needed
Complete scenario conversion	Establish a source inventory and build a new model through each workflow.	Capture coverage, mapping decisions, corrections and final configuration checks.
Controlled replication	Involve additional users, vary workflow order and record all activity intervals.	Comparable effort and error records across repeated trials.
Executable adapter	Use a supported simulator interface with explicit transformation rules.	Input-to-output tests, trace preservation and clear failure handling.
Broader application	Apply the workflow to additional scenarios and simulators.	Reuse, adaptation work and limitations of the mapping approach.
Independent result checking	Compare the reader with documented formats and application-level exports.	Agreement on active records, identities, totals and malformed-input handling.

The first priority is a complete scenario-conversion study. An independent source inventory would establish which requirements must be captured before the model is constructed. This would create a defensible denominator for source-capture coverage and allow omitted information to be distinguished from incorrectly mapped information. Starting from a new project would also make the work of selecting objects, entering values and resolving ambiguity observable. The same acceptance criteria would still be needed for both workflows, but those criteria should cover the intended configuration more fully than the selected execution conditions used here.

Repeated trials should then examine effort and maintainability. Different participants should use both workflows in varied order so that prior exposure is not consistently associated with the proposed workflow. Preparation, configuration, execution, verification and evidence handling should have explicit boundaries, with interruptions recorded separately. Corrections and repeated lookups should be recorded as they occur rather than reconstructed afterwards. Such a study could assess whether the package's initial and maintenance costs are offset by changes in later work. The present evidence does not determine the direction or size of that effect.

An executable adapter is a further development option, not an existing capability. It would require a supported means of exchanging information with the simulator, a precise interpretation of the target settings and tests for failed or incomplete transformations. The retained mapping records provide design input, but they are not an implementation specification sufficient to guarantee correctness. A useful adapter would preserve the relationship between source information and generated settings and would stop when a required target representation was ambiguous or unavailable. Its tests should compare generated configurations with independently reviewed expectations.

Additional scenarios and simulators would examine transferability. A scenario with different equipment, units or movement rules could reveal whether the current information structure is sufficiently general or merely fits the selected case. A

different simulator would test whether the separation between controlled information and product-specific mappings is useful in practice. Reuse should be assessed through the records that remain unchanged, the mappings that require revision and the work needed to establish new verification evidence. Successful transfer should not be assumed from the appearance of a general workflow diagram.

Finally, the evidence reader should be checked independently against documented file behaviour or reliable application-level exports. Tests should include active and inactive records, incomplete files, unexpected field layouts and count disagreements. This would distinguish correctness of the reported totals from correctness of the selection mechanism itself. Access arrangements would need to respect the simulator licence and the rights of the data owners.

These studies would extend the present contribution in a controlled way. The thesis provides a documented workflow and a bounded evaluation around CHESSCON; complete implementation inspection requires authorised access to the private package. Future work should retain that clarity while collecting the broader evidence needed for claims about complete conversion, sustained effort and use beyond this case.

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Appendix A

Supplementary CHESSCON Evidence

This appendix presents supplementary CHESSCON input and result details. The screens complement the comparison in Chapter 6; they do not replace the indexed result tables used for its numerical claims. Table A.1 distinguishes the contribution of each screen from the conclusions it cannot establish.

Table A.1: Meaning and limits of the supplementary CHESSCON evidence.

Evidence item	What it adds to the thesis	What it does not establish
Electric-forklift input screen	Shows the reusable `FL-1` equipment type and its visible battery settings.	It does not show how many forklift units are deployed or active.
Animated simulation screen	Shows the route, modelled areas and moving forklift symbols during execution.	It does not prove completion, accepted job totals or error-free execution.
Selected-forklift result form	Shows detailed fields available for one forklift record.	It does not represent all four forklifts or measure workflow quality.
Selected charging-station form	Shows activity and time fields for one station record.	It does not prove charging optimisation or a difference between the workflows.

The input screen defines a reusable equipment type; separate deployment records create the individual units that use that type. After execution, CHESSCON stores separate forklift and charging-station result records. The accepted comparison was calculated from preserved result tables using status-based selection and an index-count cross-check, not from these screenshots.

A.1 Electric-forklift input values

Figure A.1 shows the input definition for equipment type `FL-1`. This is a reusable type definition. The four individual forklifts used in the scenario are deployed separately from this screen.

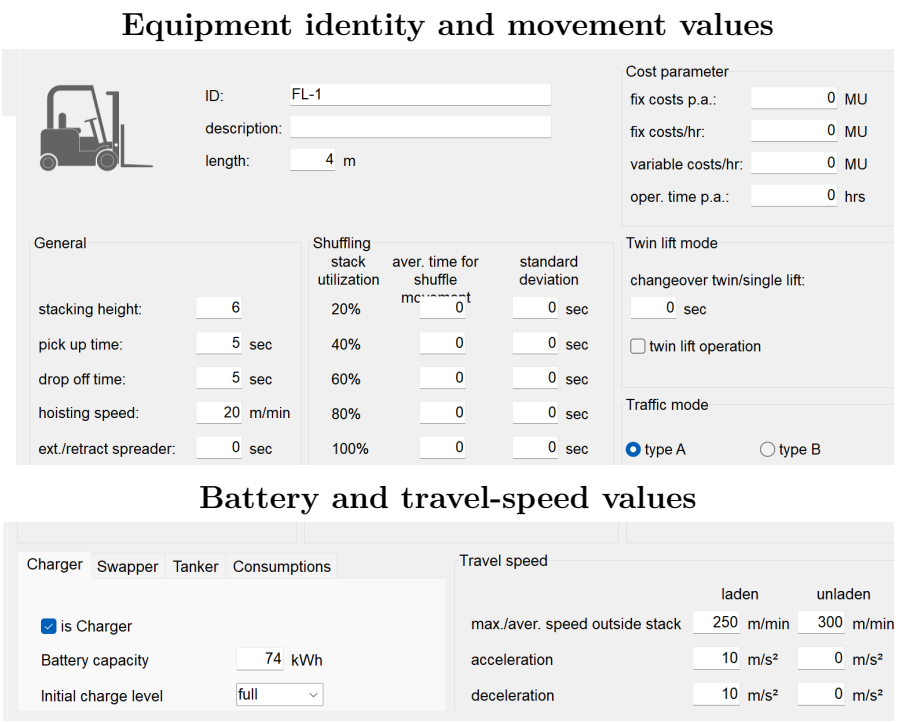


Figure A.1: CHESSCON Input view showing equipment type `FL-1`, its movement values and its visible battery settings. The option labelled "*is Charger*" is selected, battery capacity is 74 kilowatt-hours and initial charge is full. Author screenshot of CHESSCON, reproduced with permission from akquinet.

Table A.2: Visible values in the electric-forklift input screen.

Input field	Visible value	Input field	Visible value
Type identifier	`FL-1`	Forklift length	4 metres
Stacking height	6 levels	Pick-up time	5 seconds
Drop-off time	5 seconds	Hoisting speed	20 metres per minute
Maximum laden speed	250 metres per minute	Maximum unladen speed	300 metres per minute
Option labelled "*is Charger*"	Selected	Battery capacity	74 kilowatt-hours
Initial charge level	Full	Traffic mode	Type A

Table A.2 brings the relevant input values together for comparison with the scenario definition. Laden means carrying an item, and unladen means travelling without one. The screen contains additional cost and movement fields. Their visible zero values were not interpreted as study outcomes because the evaluation concerns the information workflow. The screenshot does not display the lower and upper charging thresholds, so it is not evidence for those values.

A.2 Animated simulation execution

Figure A.2 shows the model while animation is enabled. The status line reports one configured run and a current simulated time of 36 hours and 30 minutes within the 72-hour period. Two small forklift symbols can be seen moving on the connected route near the centre of the layout.

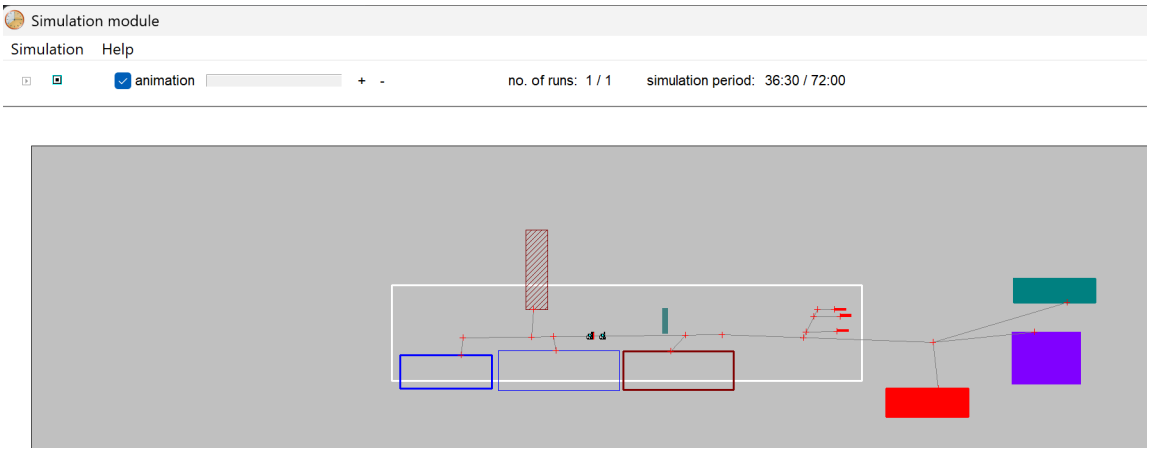


Figure A.2: CHESSCON Simulation module during an animated execution at 36:30 of the configured 72:00 hours. The moving forklift symbols provide a visual explanation of how the scenario behaves. This is an in-progress interface illustration, not the completion evidence for the controlled comparison. Author screenshot of CHESSCON, reproduced with permission from akquinet.

Table A.3: Information visible in the animated simulation screen.

Visible part	What is shown	Meaning in this study
Animation control	The animation option is selected.	CHESSCON displays object movement instead of executing only in the background.
Run counter	`1 / 1`	One simulation run is configured.
Time counter	`36:30 / 72:00`	The screenshot was taken approximately halfway through the simulated period; the execution was not yet complete.
Route network	Thin connected lines between modelled areas.	Forklifts travel along the configured route between the movement locations.
Forklift symbols	Two small moving symbols near the centre.	The image provides a visible example of equipment activity during execution.

Table A.3 relates the visible controls to the represented activity. The displayed time is elapsed simulated time, not a calendar or clock timestamp. The view does not establish the final number of completed jobs, the active result records or the absence of an execution error. Those conditions were checked after completion through the outputs described in Chapter 6.

A.3 Detailed result for one electric forklift

CHESSCON allows one forklift record to be selected from the list on the left of the result form. Figure A.3 shows the detailed values for the selected record `FL-001`. The values belong to that one record and must not be read as totals for all four forklifts.

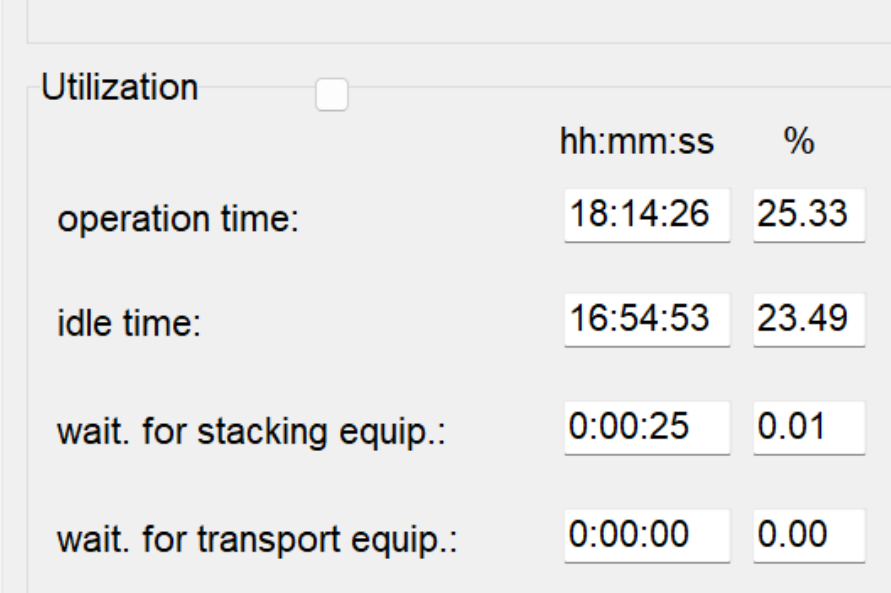


Figure A.3: Close-up of the utilisation panel for the selected forklift `FL-001`. The enlarged values show operation, idle and waiting time for this one record. Table A.4 records the additional distance, energy, charging and job values visible in the full form. It is an interface illustration with no verified accepted run identity or capture date. Its transport and stacking fields are not the accepted completed-job metric. Author screenshot of CHESSCON, reproduced with permission from akquinet.

Table A.4: Values visible for the selected forklift record.

Result field	Visible value	Result field	Visible value
Distance inside storage	51,496 metres	Laden travel outside storage	83,795 metres
Unladen travel outside storage	84,189 metres	Travel for charging	5,595 metres
Operation time	18:14:26	Idle time	16:54:53
Wait for storage equipment	00:00:25	Wait for energy station	16:04:15
Charging time	20:46:01	Charging stops	7
Total energy use	327.89 kilowatt-hours	Regenerated energy	1.79 kilowatt-hours
Energy per job	0.66 kilowatt-hours	Energy per kilometre	1.46 kilowatt-hours
Transport jobs	500	Stacking jobs	500

The time fields in Table A.4 are elapsed durations in hours, minutes and seconds, not clock timestamps. The interface uses the general label *refueling* for energy restoration. For the selected battery-electric equipment, those fields are described as charging time and stops. These values explain the form; they do not establish a workflow performance difference.

A.4 Detailed charging-station result

Figure A.4 shows the average-result form for the selected charging station `CHARGER-1`. The list on the left also contains `CHARGER-2`, confirming that CHESSCON stores a separate result record for each charging station.

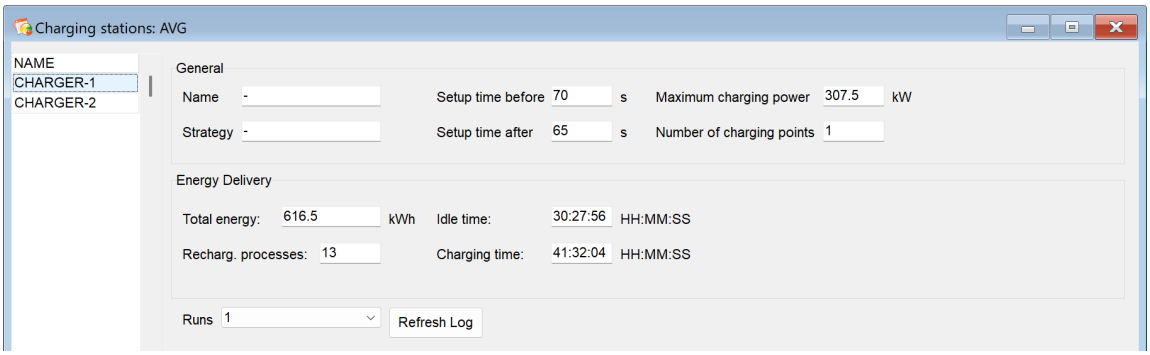


Figure A.4: Detailed CHESSCON average-result form for `CHARGER-1`. The screen reports setup times, charging capacity, delivered energy, charging processes and time use for the selected station. These are illustrative interface values rather than controlled accepted result values. Author screenshot of CHESSCON, reproduced with permission from akquinet.

Table A.5: Values visible for the selected charging-station record.

Result field	Visible value	Result field	Visible value
Selected station	`CHARGER-1`	Other station in the list	`CHARGER-2`
Setup time before charging	70 seconds	Setup time after charging	65 seconds
Maximum charging power	307.5 kilowatts	Charging points	1
Delivered energy	616.5 kilowatt-hours	Charging processes	13
Idle time	30:27:56	Charging time	41:32:04
Displayed run	1		

Table A.5 distinguishes the selected station's capacity, delivered energy and activity durations. Its time fields are elapsed hours, minutes and seconds. These illustrative values do not compare the workflows or establish an optimal number or power of stations. The controlled evaluation checked only that the two expected charging-station records were active.

Appendix B

Retrospective Literature Search Map

The literature searches used ACM Digital Library, IEEE Xplore, ScienceDirect and Google Scholar. Table B.1 maps the cited research publications to reconstructed search queries using terms from their published titles. These queries provide search aids for locating the publications. The exact original search strings and screening records were not retained, so the table does not reconstruct the original search and selection process.

Table B.1: Reconstructed search queries mapped to existing research references.

Query and focus	Search string	References mapped
Q1 — Digital-twin reviews	`("digital twin" OR "digital twins") AND ("systematic literature review" OR "systematic review" OR "umbrella review")`	[18, 31, 37, 56, 69]
Q2 — Digital-twin modelling	`("digital twin" OR "digital twins") AND (modeling OR modelling OR "virtual models" OR "multi-scenario")`	[26, 27, 37, 57, 62, 66, 70]
Q3 — Manufacturing and scenario applications	`("digital twin" OR "digital twins") AND (manufacturing OR "shop floor" OR "shop-floor" OR "aircraft assembly")`	[27, 34, 39, 40, 51, 70, 73]
Q4 — Digital-twin frameworks and architecture	`("digital twin" OR "digital twins") AND (framework OR frameworks OR architecture)`	[1, 18, 19, 25, 32, 45, 65, 70]
Q5 — Digital-twin design and testbeds	`("digital twin" OR "digital twins") AND ("product design" OR "system design" OR "product-service systems" OR testbed)`	[8, 38, 66, 67]
Q6 — Digital-twin integration and coupling	`("digital twin" OR "digital twins") AND ("collaborative network" OR "virtual-real integration" OR "data-model coupling")`	[9, 28, 41]
Q7 — Cognitive digital-twin background	`"cognitive digital twin" AND (vision OR challenges OR opportunities)`	[71]

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Table B.1 (continued)

Query and focus	Search string	References mapped
Q8 — Software traceability techniques	`"software traceability" AND ("machine learning" OR "natural language processing" OR barriers)`	[5, 49, 54]
Q9 — Traceability practice and requirements	`traceability AND ("industrial experience" OR "Agile development" OR "traceability practices" OR "pre-requirements specification")`	[3, 17, 42, 48]
Q10 — Traceability links and artefacts	`traceability AND ("execution traces" OR BPMN OR "test-to-code" OR "issues and commits" OR "artifact traceability graphs")`	[11,33,68,72]
Q11 — Interoperability	`interoperability AND ("digital twin" OR "digital twins" OR simulation OR "data exchange")`	[2, 35, 59]
Q12 — Consistency management	`consistency AND ("model-based development" OR "collaborative engineering" OR "language specifications")`	[29, 36, 64]
Q13 — Model transformation and reuse	`"model transformation" AND (incremental OR traceability OR reuse)`	[23, 44]
Q14 — Software configuration	`configuration AND ("product lines" OR "software systems" OR "software architecture")`	[6, 10, 12, 16]
Q15 — Provenance	`provenance AND (software OR "source code" OR reproducibility)`	[14, 55, 61]
Q16 — Reproducibility	`reproducibility AND ("software engineering" OR "scientific software" OR "Jupyter notebooks" OR "stack overflow" OR "machine learning")`	[13, 20, 21, 46, 52]
Q17 — Simulation verification and credibility	`("verification and validation" AND (simulation OR simulations)) OR ("credibility evaluation" AND "digital twin")`	[15, 58, 59]
Q18 — Design Science Research	`"design science" AND ("information systems" OR reliability OR transparency)`	[22,47,50,60]
Q19 — Maritime and port architecture	`(maritime OR "smart port") AND (architecture OR "digital platform")`	[19, 30, 43]

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Table B.1 (continued)

Query and focus	**Search string**	**References mapped**
Q20 — Conceptual modelling and simulation input data	`("conceptual modelling" AND simulation) OR ("input data management" AND "discrete event simulation")`	[7, 53]

The mapping covers all 70 research publications in the bibliography. CHESSCON product information, ISO 23247-1:2021 and the TIC4.0 standards and definitions portal remain separate authoritative sources [4, 24, 63].



Faculty of Computing, Blekinge Institute of Technology, 371 79 Karlskrona, Sweden